

TOPIC 6. ELECTRICAL PROPERTIES OF MATERIALS

Band structure and electrical properties

Conducting materials

Metallic conductors

Non-metallic conductors

Semiconducting materials

Intrinsic

Extrinsic (n and p)

Insulating and Dielectric materials

Learning objectives:

- Know the classification of the materials according to their electrical and magnetic properties
- Understand how the structure and composition of a material determine its response to a electrical and to a magnetic field.
- Understand the influence of the temperature on the electrical and magnetic properties of a material.

BAND STRUCTURES OF SOLIDS

As atoms come within close proximity → orbitals split into **energy bands**

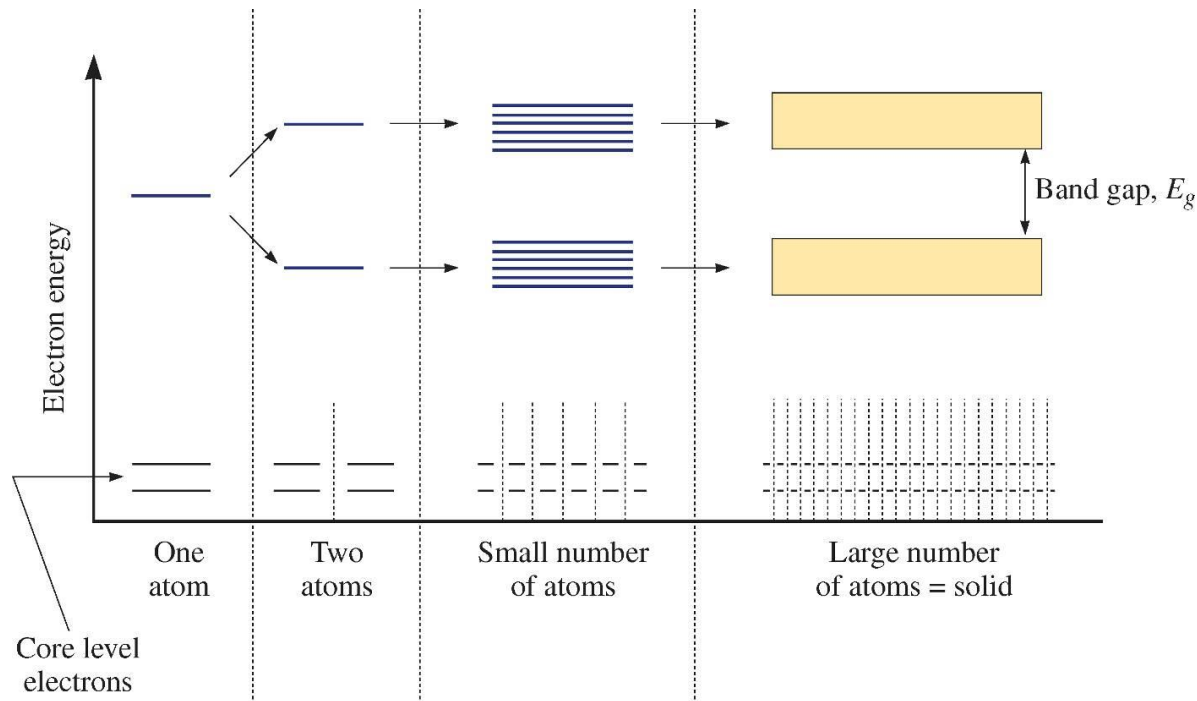
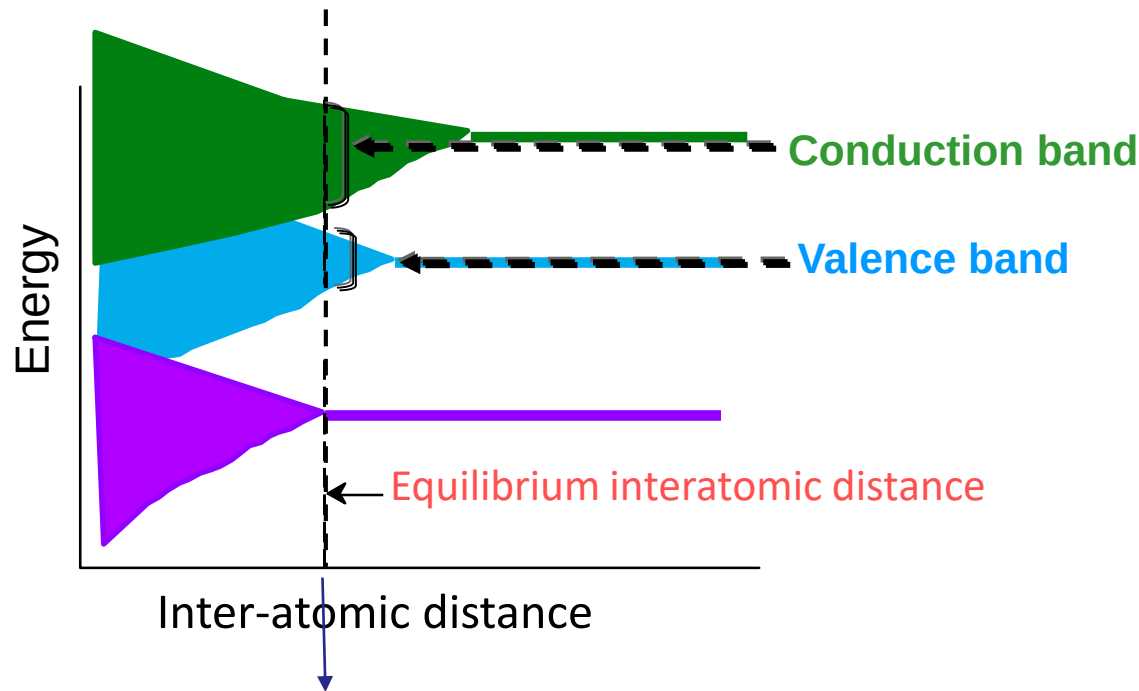


Figure 19-2 The energy levels broaden into bands as the number of electrons grouped together increases. (Courtesy of John Bravman.)

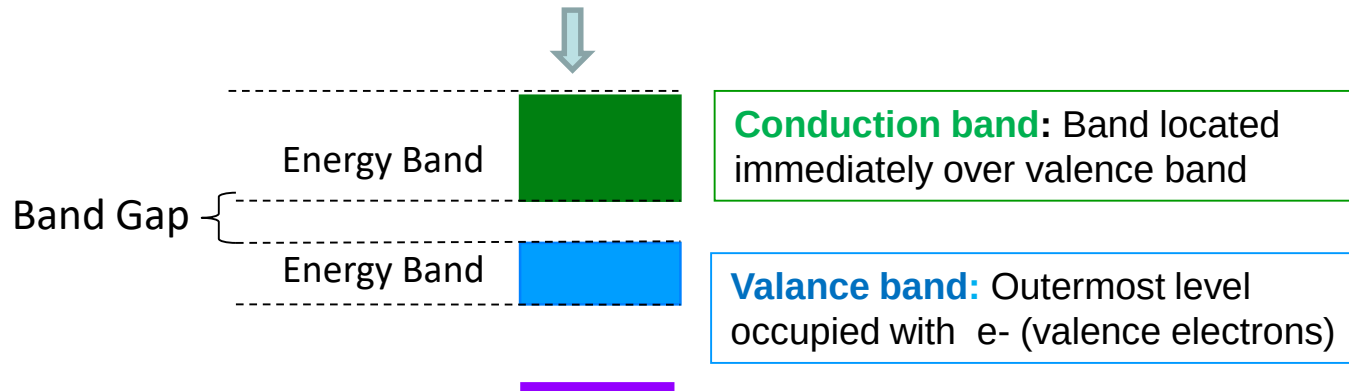
Electron energy versus interatomic separation for an “aggregate” of atoms

Upon close approach, each of the energy atomic states splits to form an electron energy band consisting of states with so close energies that they form a continuous energy band.

BAND STRUCTURES OF SOLIDS



The electron energy **band structure** for a solid is given by its interatomic separation.



BAND STRUCTURES OF SOLIDS

Classification of the materials according to their electrical properties

INSULATORS	SEMICONDUCTORS	CONDUCTORS
$\sigma \cong 10^{-10} - 10^{-20} (\Omega \cdot m)^{-1}$	$\sigma \cong 10^4 - 10^{-6} (\Omega \cdot m)^{-1}$	$\sigma \cong 10^7 (\Omega \cdot m)^{-1}$
e.g.: <i>Polyethylene</i>	e.g.: <i>Si, Ge</i>	e.g.: <i>Metals</i>



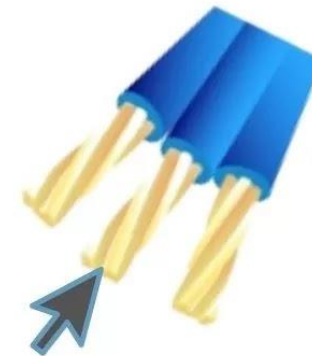
PVC coating

Insulator



Microprocessors

Semiconductor

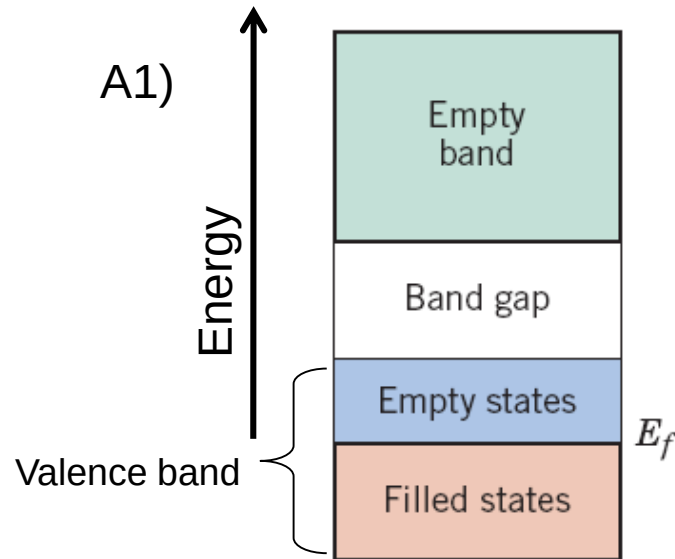


Copper (Cu) wire

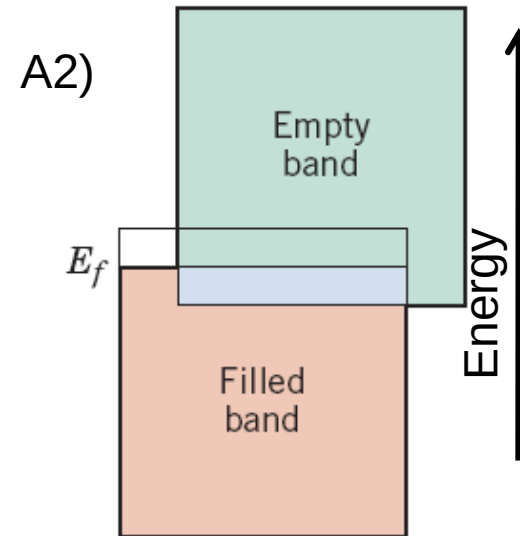
Conductor

BAND STRUCTURES IN SOLIDS

Band structure of conductors



The Valance band is half-filled since there is only one s electron (Na, Cu, Al)

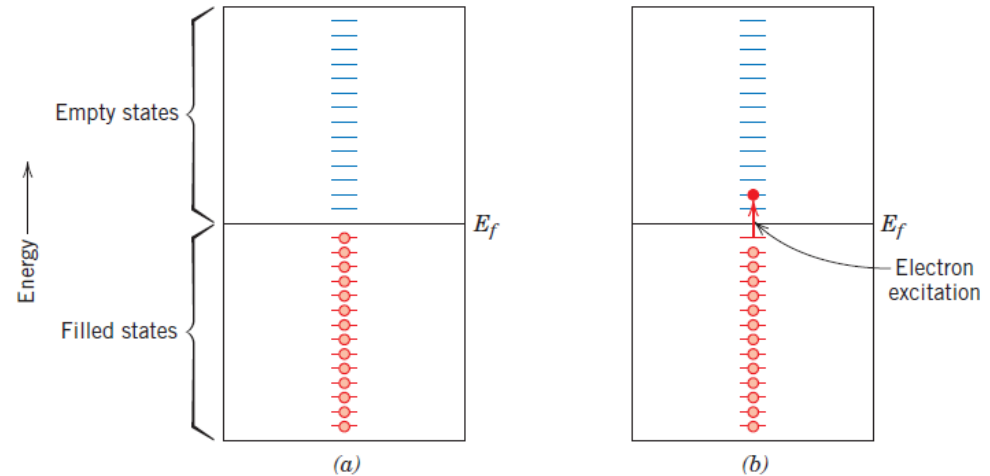
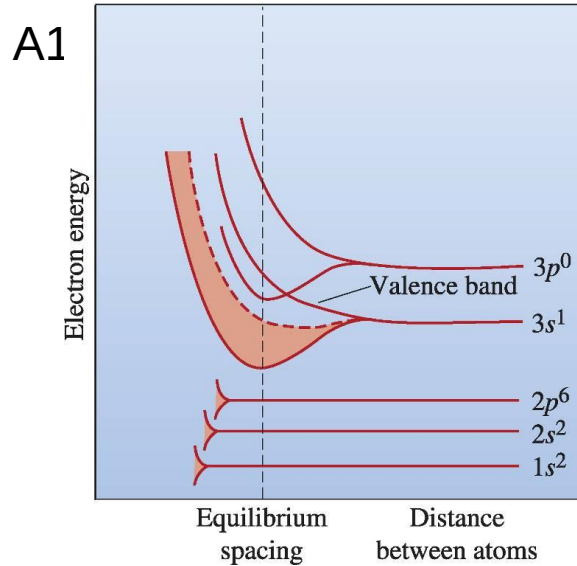


Overlapping of filled 3s valancea band and empty 3p conduction bands (Mg).

Atomic Number	Element	Electronic Configuration							
		1s	2s	2p	3s	3p	3d	4s	4p
12	Mg	2	2	6	2				
13	Al	2	2	6	2	1			
29	Cu	2	2	6	2	6	10	1	

BAND STRUCTURES IN SOLIDS

Band structure of conductors



For a metal, occupancy of electron states
(a) before and (b) after an electron excitation.

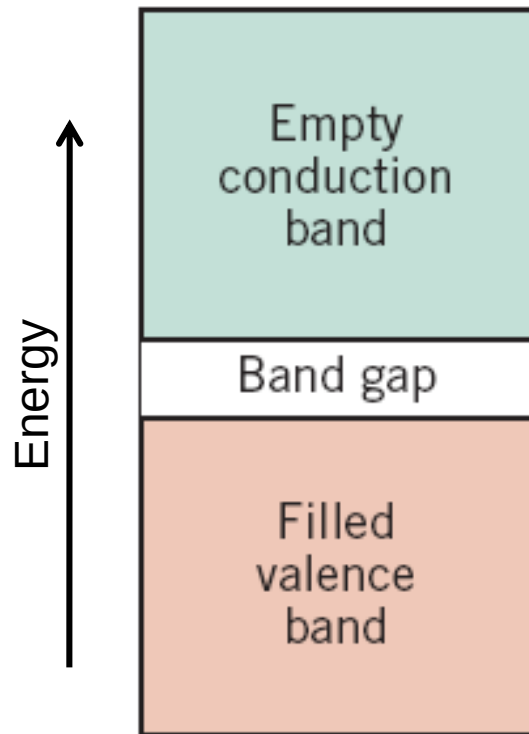
The band is half-filled since there is only one s electron (Na, Al, Cu)

CONDUCTORS

Atomic Number	Element	Electronic Configuration							
		1s	2s	2p	3s	3p	3d	4s	4p
12	Mg	2	2	6	2				
13	Al	2	2	6	2	1			
29	Cu	2	2	6	2	6	10	1	

BAND STRUCTURES IN SOLIDS

Band structure of semiconductors

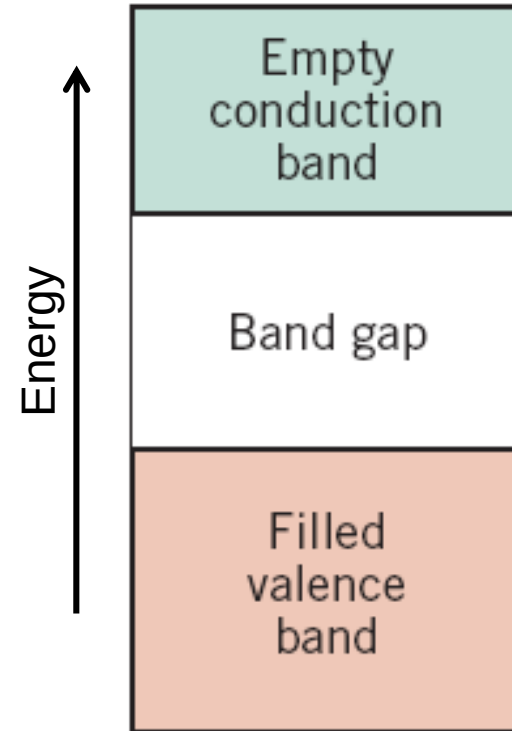


The band structure in semiconductors is relatively **narrow** (< 2 eV).

(Pure silicon : $E_g = 1.1$ eV)

Pure germanium: $E_g = 0.67$ eV)

Band structure of insulators



The filled valence band is separated from the empty conduction band by a relatively **large band gap**, E_g (> 2 eV)

(Pure diamond : $E_g = 6-7$ eV)

TOPIC 6. ELECTRICAL PROPERTIES OF MATERIALS

Band structure and electrical properties

Conducting materials

Metallic conductors

Non-metallic conductors

Semiconducting materials

Intrinsic

Extrinsic (n and p)

Insulating and Dielectric materials

METALLIC CONDUCTORS

$$I = \frac{V}{R}$$

Ionic conductivity

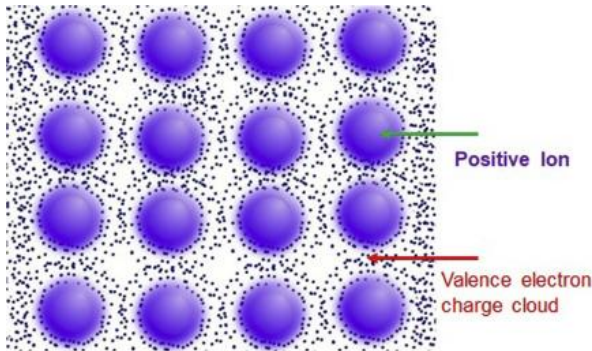
Aqueous electrolytes,
Some ceramics, etc

Electrical conductivity

Metals

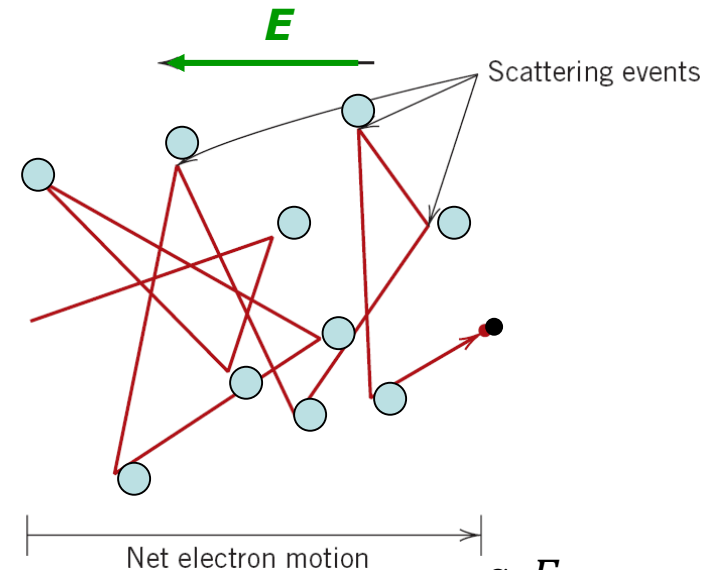
I = electric current, A (amperes)
 V = potential difference, V (volts)
 R = resistance, Ω (ohms)

In a conductive metal it can be said that there is a cloud of electrons around the nuclei



E = uniform electric field intensity (V/m²)
 q_e = electron charge (1.6×10^{-19} C)
 m_e = mass of electron (9.109×10^{-31} Kg)
 a = acceleration (m/s²)
 F = Force(N)

When an electric field E is applied: electrons **accelerate**



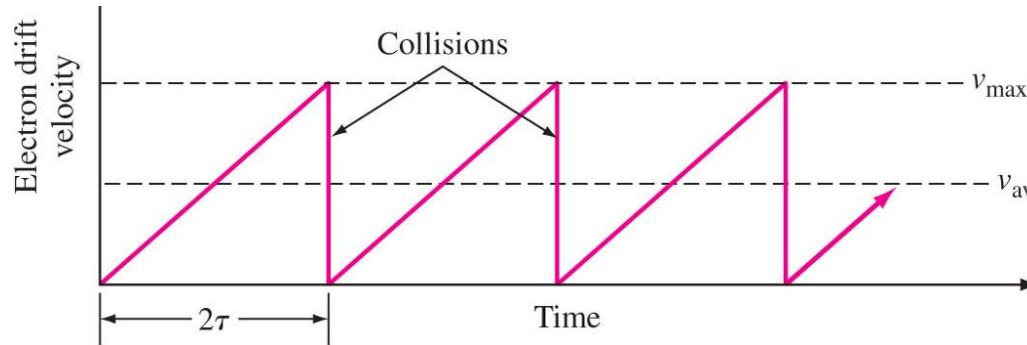
$$F = q_e E = m_e \cdot a \Rightarrow a = \frac{q_e E}{m_e}$$

METALLIC CONDUCTORS

ORIGIN OF METAL'S RESISTIVITY

Mobility of electrons, μ

however, for a given $E \Rightarrow I$ constant $\Rightarrow$ speed of e^- : constant



Electron **drift velocity** versus time for classical model for electrical conductivity of a free electron in a metal

There are “friction forces”, due to **collisions**, that counteract E acceleration $\rightarrow$ **resistivity**

Average drift velocity v_d :

$$V_d = \frac{\tau \cdot |q_e| \cdot E}{m_e}$$

$$V_d = \mu \cdot E$$

Drift velocity v : average speed of an electron in the direction of the force generated by the field

E = uniform electric field intensity (V/m^2)

q_e = electron charge ($1.6 \times 10^{-19} C$)

τ = Relaxation time

m_e = mass of electron ($9.109 \times 10^{-31} Kg$)

where

$$\mu = \frac{\tau \cdot |q_e|}{m_e}$$

μ = electron mobility ($m^2/(V \cdot s)$)

METALLIC CONDUCTORS

ELECTRONIC MOBILITY Mobility of electrons, μ

Average drift velocity v_d :

$$V_d = \frac{\tau \cdot |q_e| \cdot E}{m_e}$$

$$V_d = \mu \cdot E$$

E = uniform electric field intensity (A/m^2)
 q_e = electron charge ($1.6 \times 10^{-19}C$)
 τ = Relaxation time
 m_e = mass of electron ($9.109 \times 10^{-31} Kg$)
 n = number of free conducting electrons

where

$$\mu = \frac{\tau \cdot |q_e|}{m_e}$$

μ = electron mobility ($m^2/(V \cdot s)$)

Considering a material with a current density of J (A/m^2):

$$\text{Current density} = J = n \cdot |q_e| \cdot V_d = n \cdot |q_e| \cdot \mu \cdot E \quad \text{since } J = \sigma E$$

$$\sigma = \frac{J}{E} = n \cdot |q_e| \cdot \mu$$

METALLIC CONDUCTORS

ELECTRONIC MOBILITY

Mobility of electrons, μ

Considering a material with a current density of J (A/m²):

$$\text{Current density} = J = n \cdot |q_e| \cdot V_d = n \cdot |q_e| \cdot \mu \cdot E$$

since $J = \sigma E$

$$\sigma = \frac{J}{E} = n \cdot |q_e| \cdot \mu$$

σ = electrical conductivity ($\Omega \cdot \text{m}$)⁻¹

ρ = electrical resistivity ($\Omega \cdot \text{m}$)

E = uniform electric field intensity (V/m²)

q_e = electron charge ($1.6 \times 10^{-19} \text{C}$)

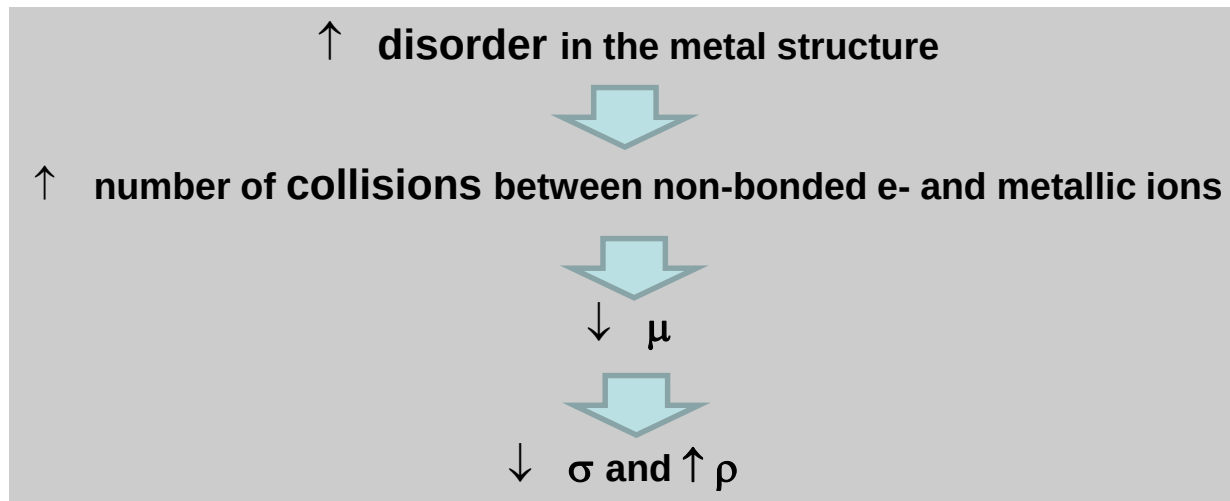
τ = Relaxation time

m_e = mass of electron ($9.109 \times 10^{-31} \text{Kg}$)

n = number of charge carriers

In metals: non-bonded e-

Metal's Conductivity depends on the number of electrons per volume unit AND THEIR MOBILITY



ELECTRICAL RESISTIVITY OF METALS

Matthiessen law

$$\rho_{total} = \rho_T + \rho_r$$

Residual resistivity

Thermal resistivity

Thermal resistivity

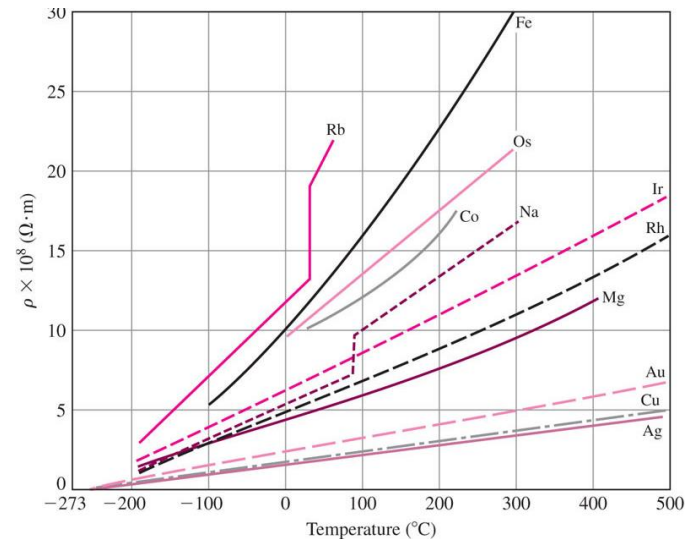
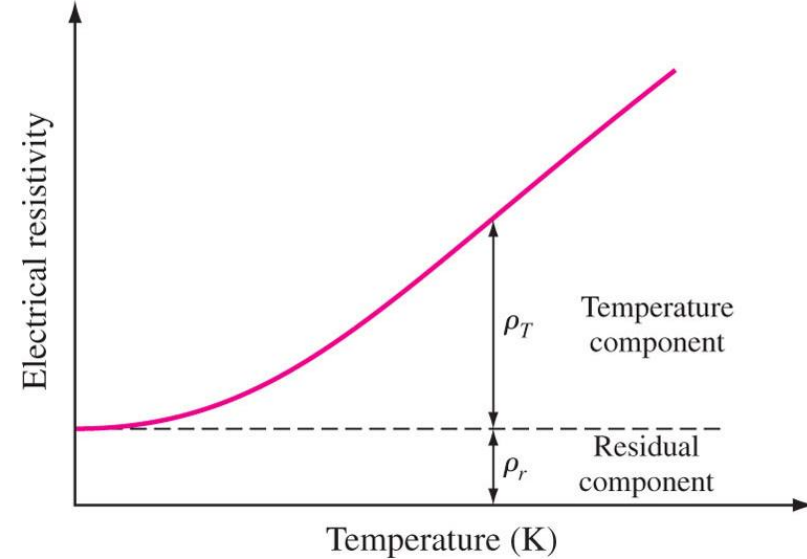
$$\rho_T = \rho_0(1 + \alpha \cdot \Delta T)$$

ρ_0 = resistivity at 0°C ($\Omega \cdot m$)

α_T = temperature coefficient of resistivity, ($^{\circ}C^{-1}$)

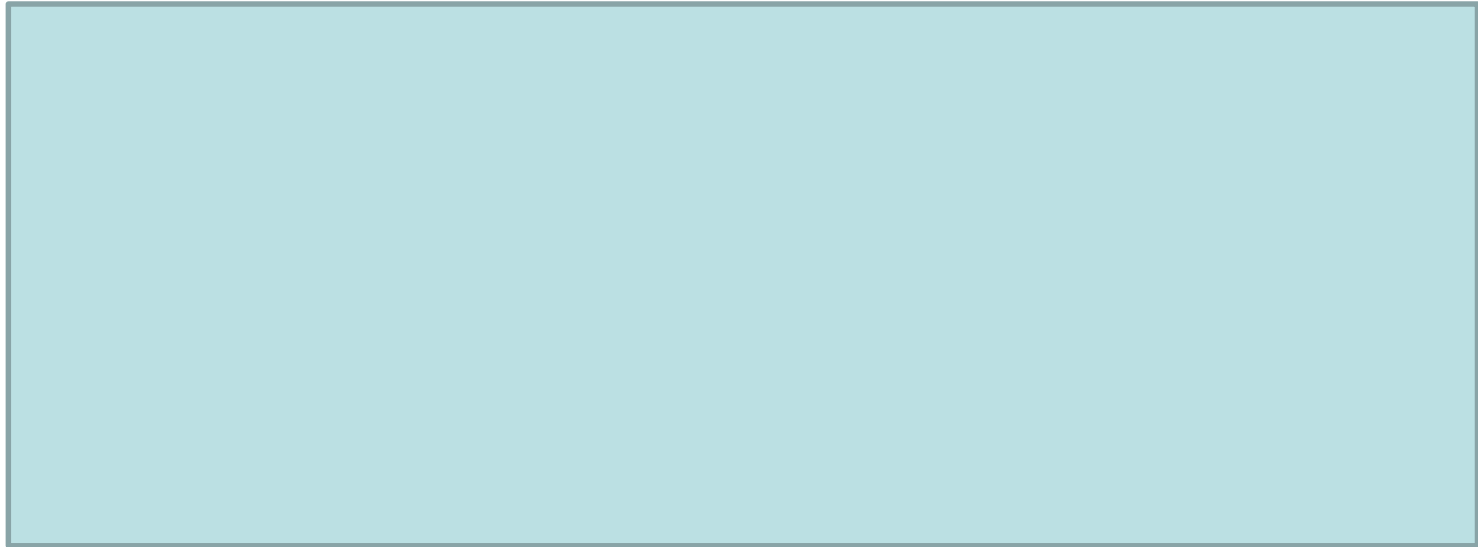
Metal	Electrical resistivity at 0°C ($\mu\Omega \cdot cm$)	Temperature resistivity coefficient α_T ($^{\circ}C^{-1}$)
Aluminum	2.7	0.0039
Copper	1.6	0.0039
Gold	2.3	0.0034
Iron	9	0.0045
Silver	1.47	0.0038

Temperature Resistivity Coefficients for selected materials



The effect of temperature on the electrical resistivity of selected metals

Q1. At what temperature will a silver wire exhibit the same resistivity as an aluminium wire at 45 °C. The resistivities of silver and aluminium at 0 °C are $1.47 \cdot 10^{-6}$ and $2.70 \cdot 10^{-6} \Omega \cdot \text{cm}$ respectively and their resistivity coefficients, 0.0038 and 0.0039 °C⁻¹, respectively.

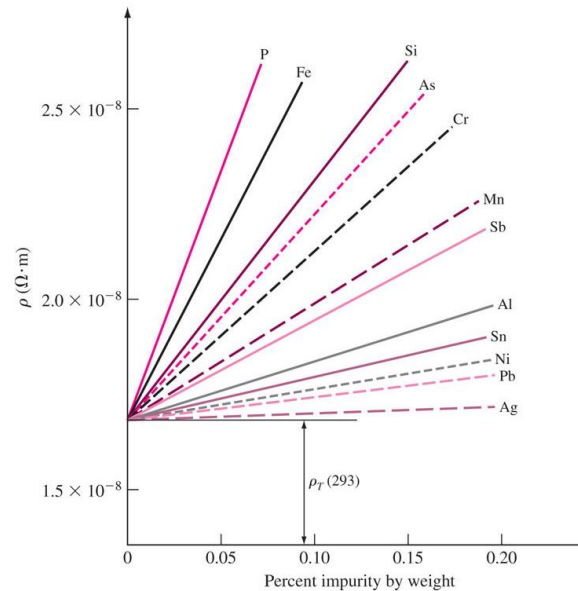
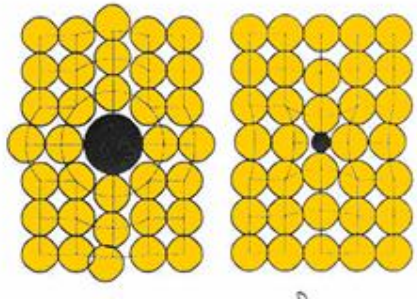


ELECTRICAL RESISTIVITY OF METALS

RESIDUAL RESISTIVITY

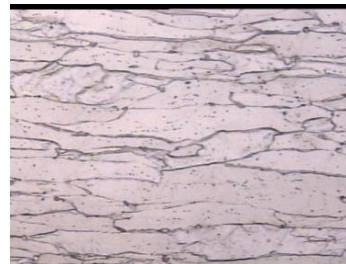
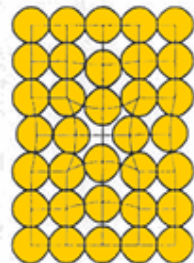
ρ_r depends on:

- Concentration and type of impurities or alloying elements (impurity resistivity, ρ_i)



Effect of small additions of various elements on the room temperature electrical resistivity of copper

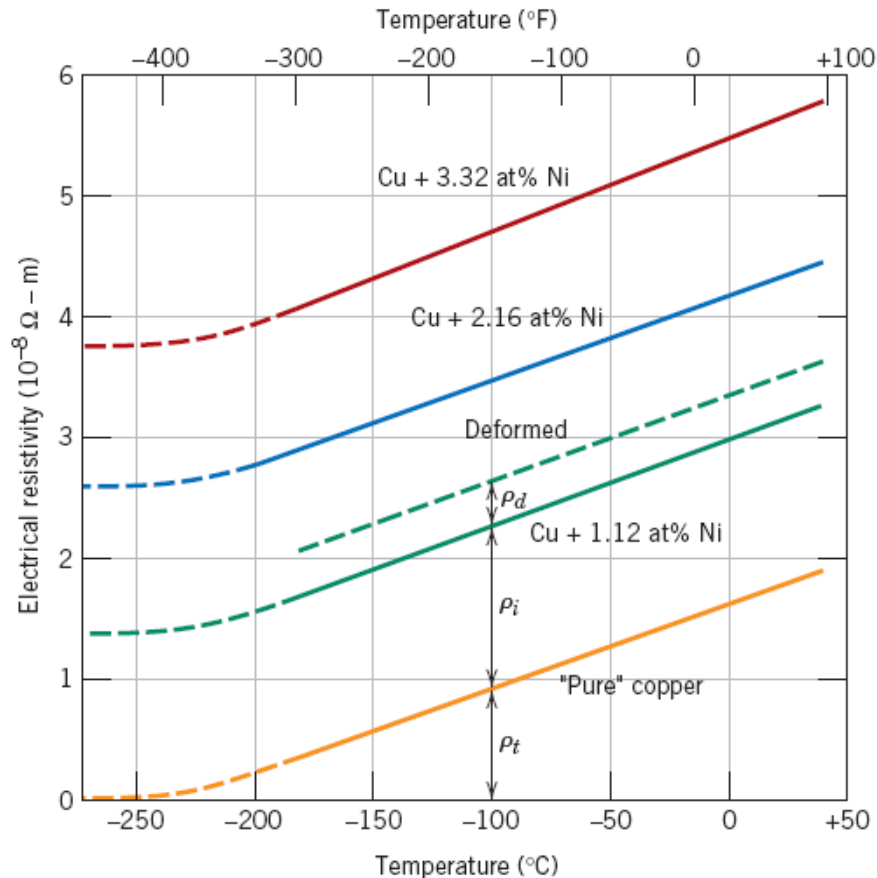
- Structural imperfections (deformation resistivity, ρ_d): dislocations, grain boundaries, ...



ρ_r increases with “the degree of distortion” of metal net

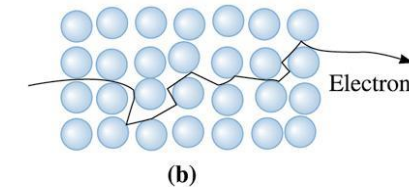
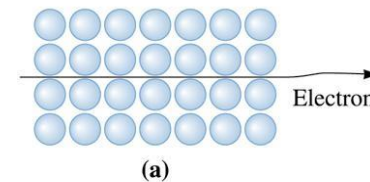
ELECTRICAL RESISTIVITY OF METALS

RESIDUAL RESISTIVITY

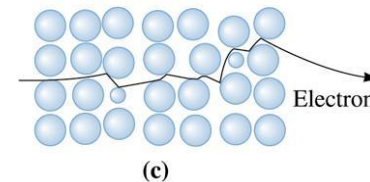


Movement of an electron through

perfect crystal



crystal heated at high temperature



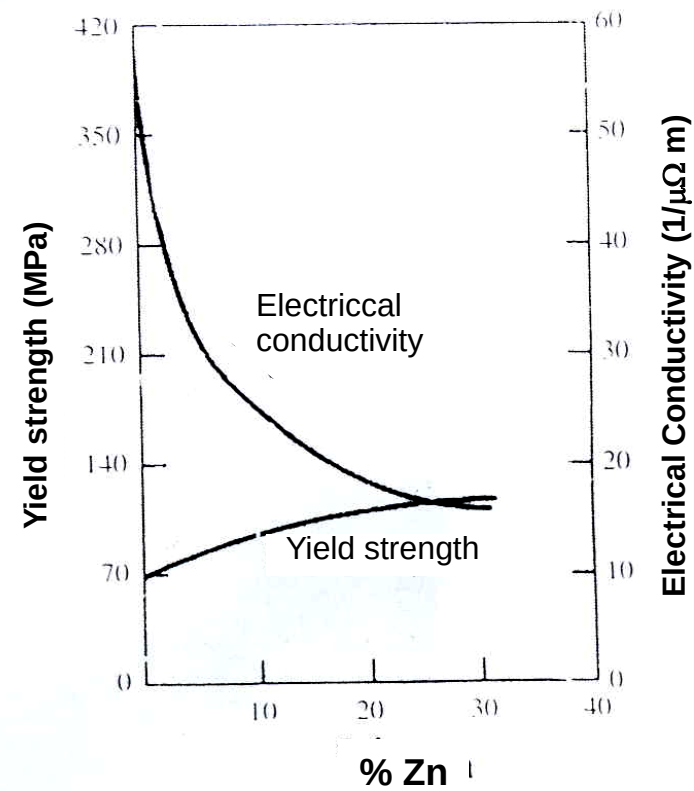
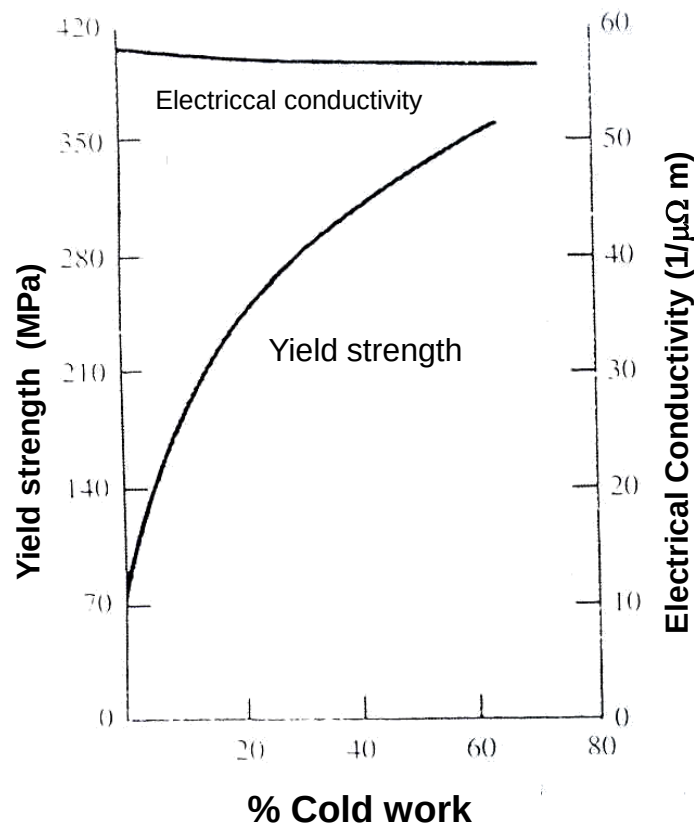
crystal containing atomic level defects

Scattering of the electrons reduces the mobility and conductivity

ELECTRICAL RESISTIVITY OF METALS

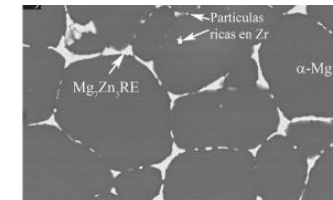
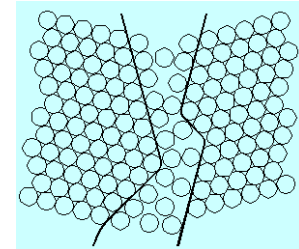
Mechanical properties and resistivity

All hardening methods cause, to a greater or lesser extent, $\downarrow \sigma$

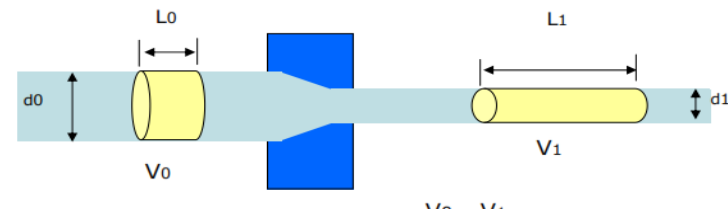


Q2. Reason what effect the following strategies would have on the conductivity and tensile strength of an aluminum alloy wire:

- Alloying with a more conductive metal as copper.
- Decrease on its working temperature.
- Thermal treatment that increases the grain size of the alloy.
- Treatment that causes intermetallic precipitation.
- Metal drawing (forming) that involves a high % of cold work.



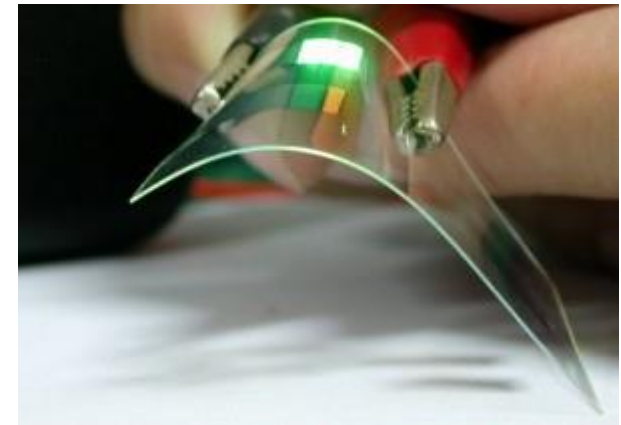
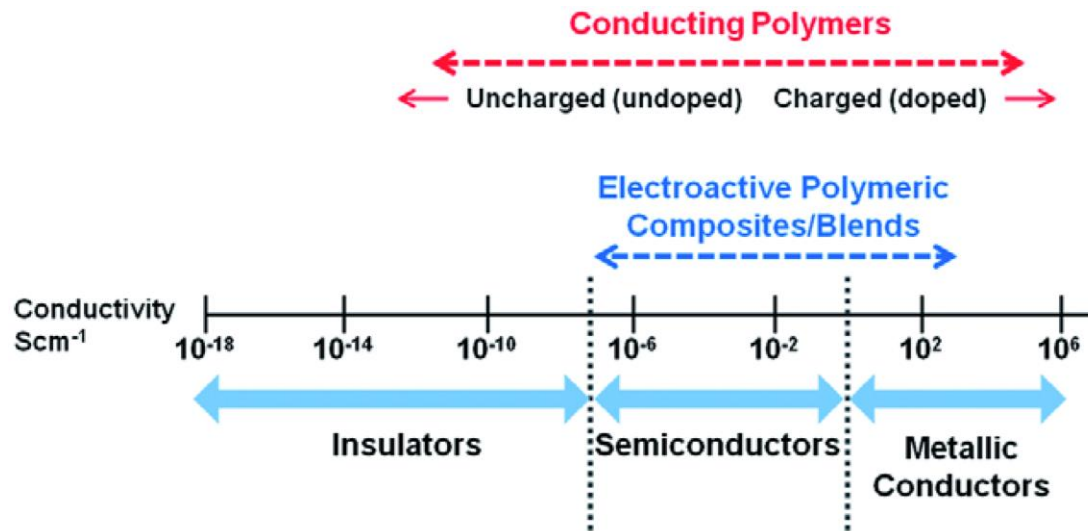
Indicate which would be the best strategy to reduce the weight of the wire while maintaining its tensile strength and conductivity.



NON- METALLIC CONDUCTORS

Conductive polymers

A few thermoset polymers, when they are specifically manufactured, can become conductive



Advantages against metals:
flexibility and easy formability

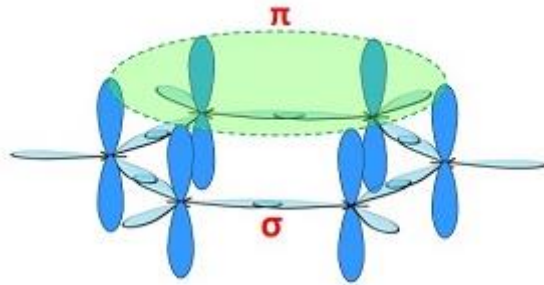
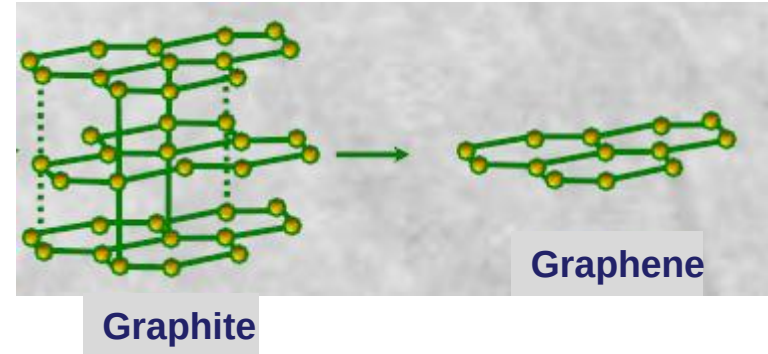
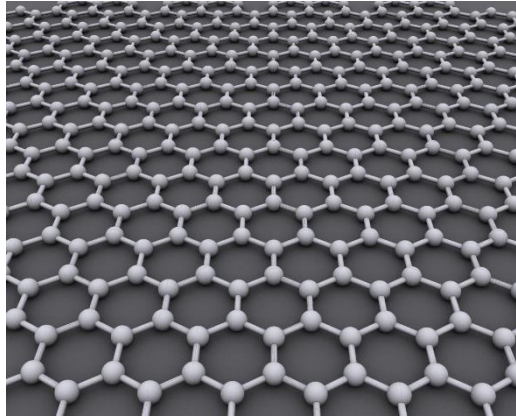
Main mechanism for transmitting electric charge:

- Delocalized e^- movement
- e^- exchange reaction between neighboring redox sites

NON- METALLIC CONDUCTORS

Graphene

Single layer
hexagonal lattice
formed by hexagons
with an atom in each
vertex



Huge π cloud with delocalized e^-



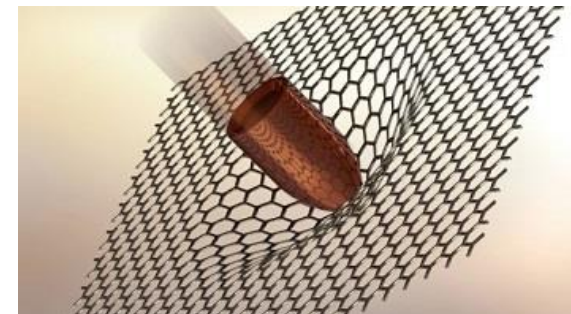
e^- can move all over the surface at a speed higher than in any other material

$$(\approx 1000 \text{ km/s})$$

$$\rho(\text{graphene}) = 0.96 \cdot 10^8 (\Omega \cdot \text{m})^{-1}$$

$$\rho(\text{Cu}) = 0.60 \cdot 10^8 (\Omega \cdot \text{m})^{-1}$$

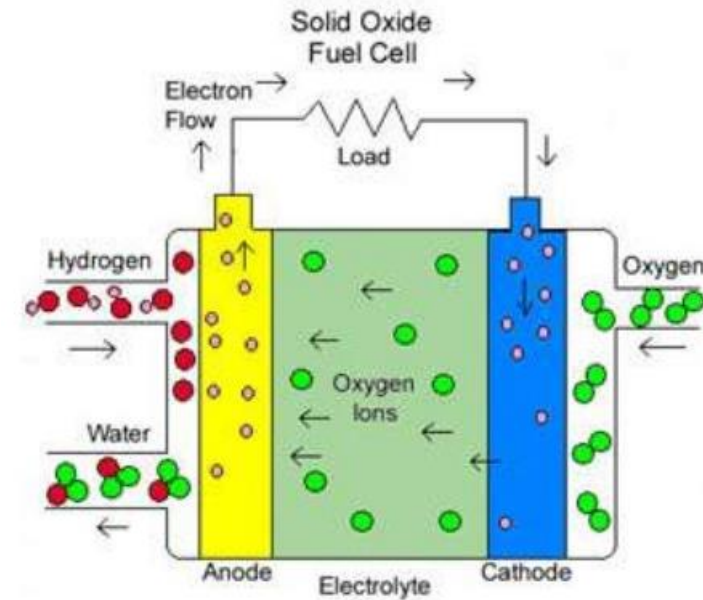
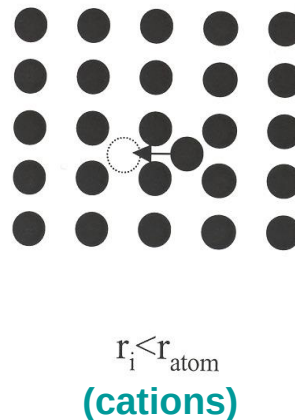
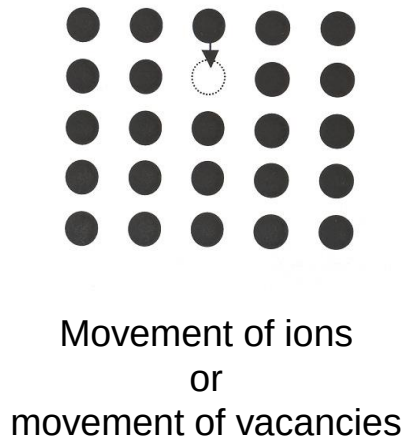
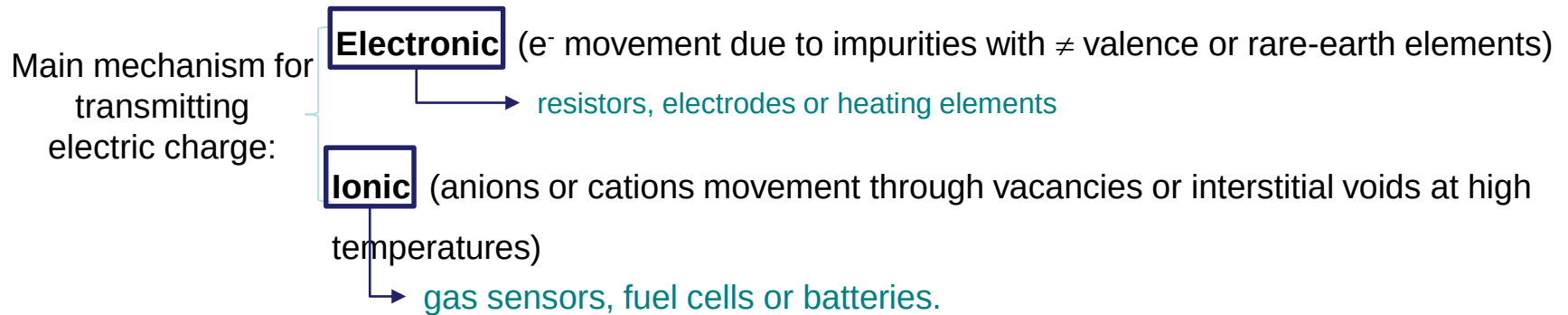
$$\rho(\text{Si}) = 4.5 \cdot 10^{-4} (\Omega \cdot \text{m})^{-1}$$



NON- METALLIC CONDUCTORS

Conductive ceramics

Ceramics are traditionally insulators, but there are some advanced ceramics that are good electrical conductors



Q3. Compare the effect of following factors on the conductivity of an ionic conductor and a metallic conductor

	σ - Ionic ceramic	σ - metal
Vacant positions in the net		
Cations in interstitial positions		
Anions in interstitial positions		
Increase of temperature		
Disorder of the net		
Doping with substances with other valences		

TOPIC 6. ELECTRICAL PROPERTIES OF MATERIALS

Band structure and electrical properties

Conducting materials

Metallic conductors

Non-metallic conductors

Semiconducting materials

Intrinsic

Extrinsic (n and p)

Insulating and Dielectric materials

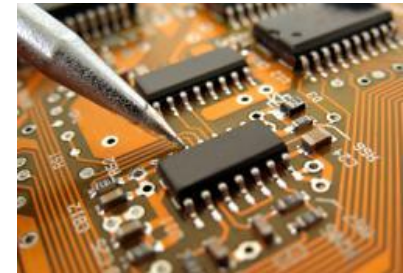
SEMICONDUCTORS

- Semiconductors are materials whose
 - Electrical conductivity is approximately midway between the values for good conductors and insulators or definition according to energy-band model
 - Electronic band structure consists of a full valence band and an empty conduction band both separated by a energy gap $E_g < 2\text{eV}$.

20th century: manufacturing of materials with extremely high chemical and structural purities



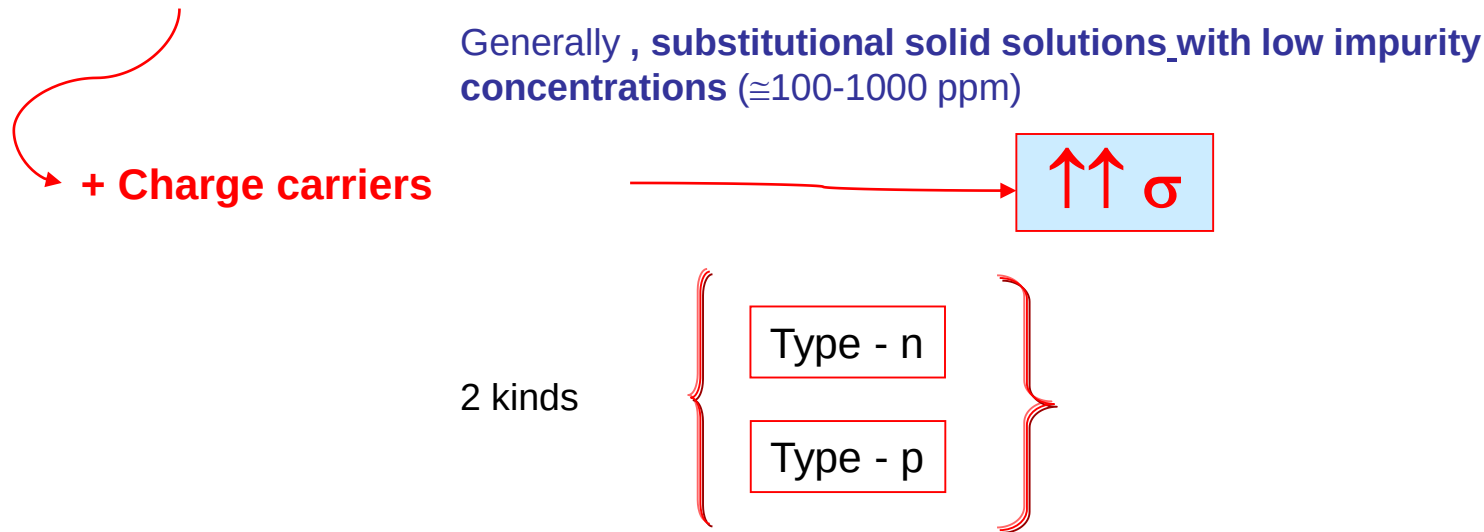
Mediocre but
EASILY CONTROLLABLE
conductivities



TYPES OF SEMICONDUCTORS DEPENDING ON THEIR BAND STRUCTURE

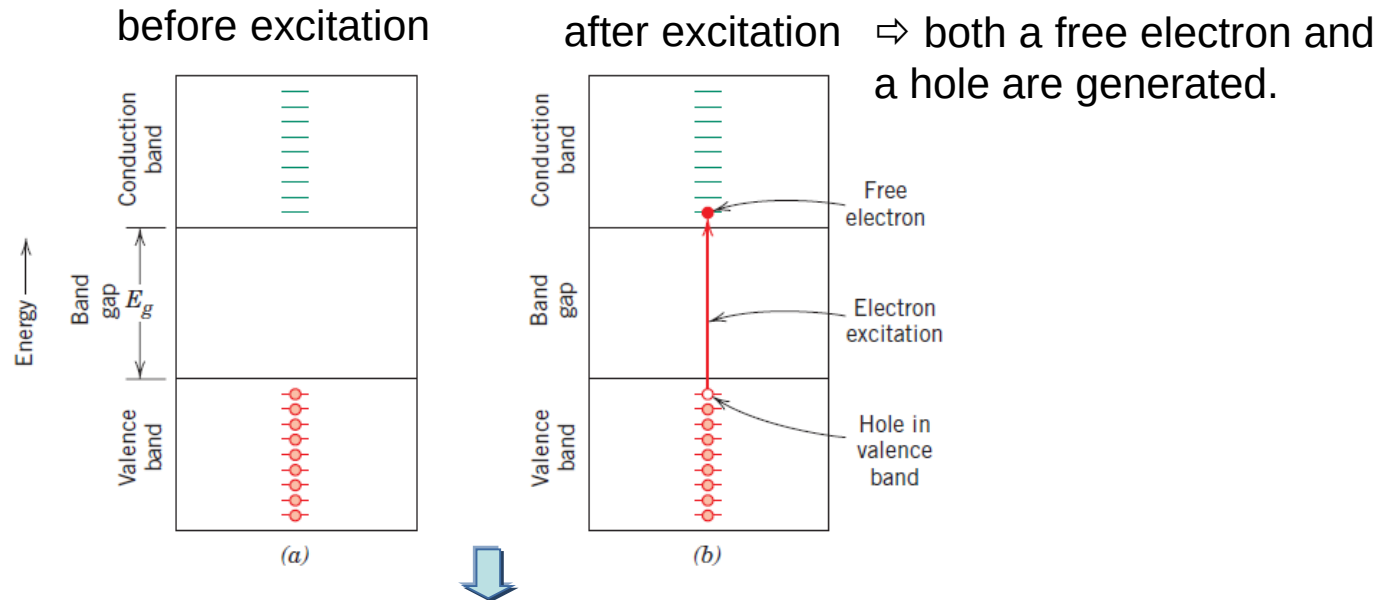
Intrinsic Semiconductors $\Rightarrow$ Their electrical conductivity is determined by their inherent conductive properties of the pure material (e.g. elemental silicon and germanium)

Extrinsic Semiconductors $\Rightarrow$ The electrical properties are given by adding dopants with a number of electrons different from the solvent



INTRINSIC SEMICONDUCTORS

Electrical or thermal excitation causes an electron to pass **from the valence band to the conduction band**, this causes the **generation of a hole** in the valence band. The conduction occurs by movement of both the electrons and the holes.



2 types of charge carriers

electrons (n) in Conduction Band
($q = 1,6 \cdot 10^{-19} \text{ C}$)

Holes (p) in valence band
($q = +1,6 \cdot 10^{-19} \text{ C}$)

$$\sigma = n \cdot |q_e| \cdot \mu_e + p \cdot |q_e| \cdot \mu_h = n \cdot |q_e| \cdot (\mu_e + \mu_h)$$

INTRINSIC SEMICONDUCTORS

Mechanism

- Bonding valence electron unable to move unless sufficient energy is provided to excite them from their bonding position.
- A critical amount of energy is supplied to a valence electron to excite it away from its bonding position
- It becomes a free conduction electron and leaves behind a positively charged hole in the crystal lattice

$$\sigma = n \cdot |q_e| \cdot \mu_e + p \cdot |q_h| \cdot \mu_h = n \cdot |q_e| \cdot (\mu_e + \mu_h)$$

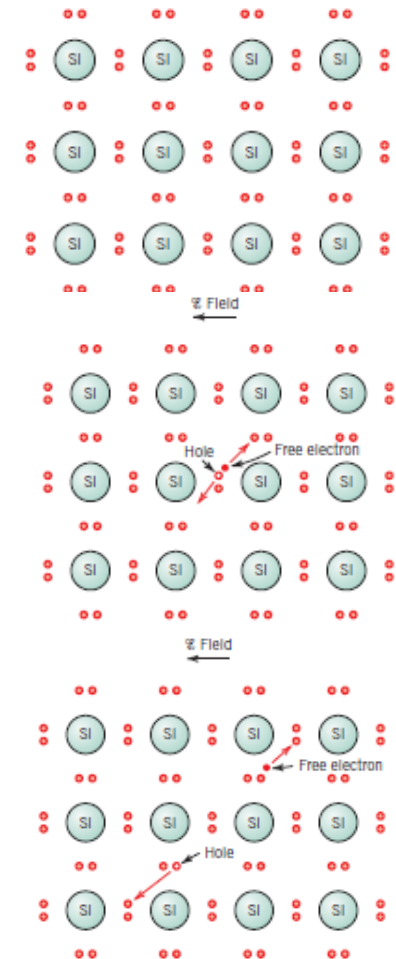
$$\text{Given : } n = p \text{ and } |q_e| = |q_h|$$

n_e : number of conduction electrons per unit volume

n_h : number of conduction holes per unit volume

$|q_e|$: absolute value of electron or hole charge

μ_e = e- mobility ; μ_h = hole mobility



INTRINSIC SEMICONDUCTORS

EFFECT OF TEMPERATURE IN INTRINSIC SEMICONDUCTORS

For intrinsic semiconductors: **n** and **p** increase with temperature

$$n_i \propto \exp\left(-\frac{E_g}{2kT}\right)$$

The temperature tends to disorder the crystalline structure, but this effect is masked by the increase on the number of charge carriers

since: $\sigma \propto n_i$

$$\sigma = \sigma_0 \exp\left(-\frac{E_g}{2kT}\right)$$

$$\ln \sigma = \ln \sigma_0 - \frac{E_g}{2kT}$$

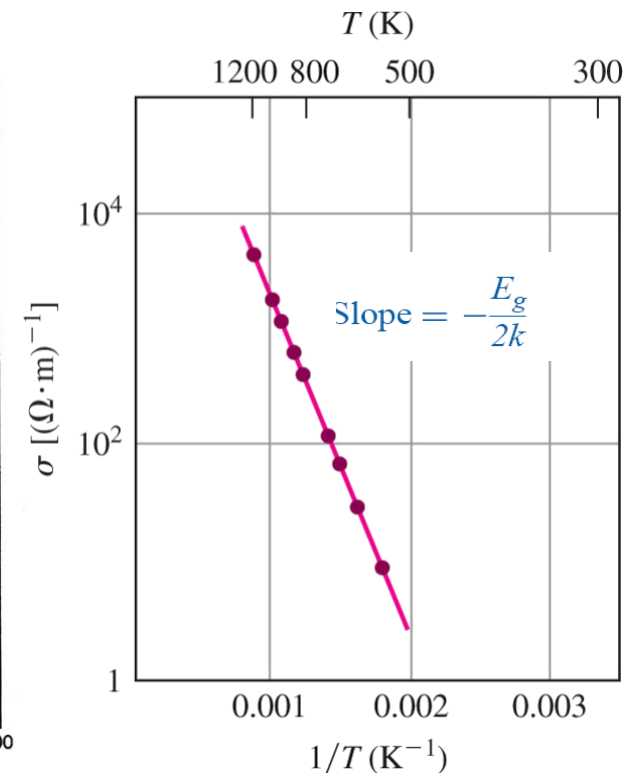
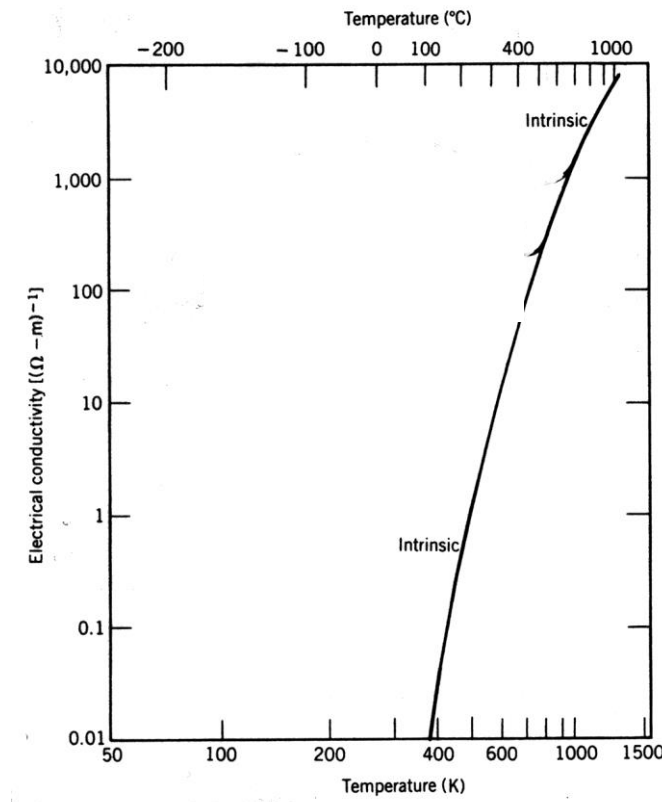
$$\ln \sigma = C' - E_g/(2k_B T)$$

k_b = Boltzmann's constant

T = temperature in K

E_g = energy band gap

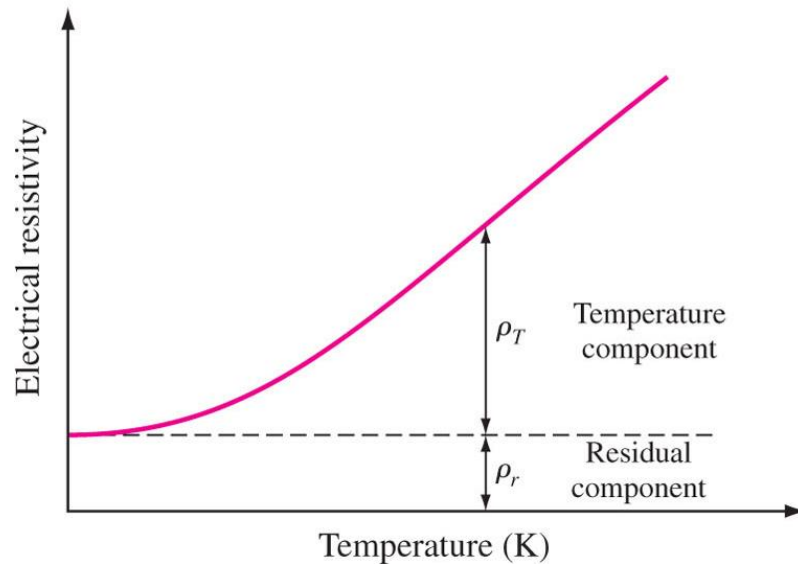
$n_i = p_i$ = number of excited e^- /holes



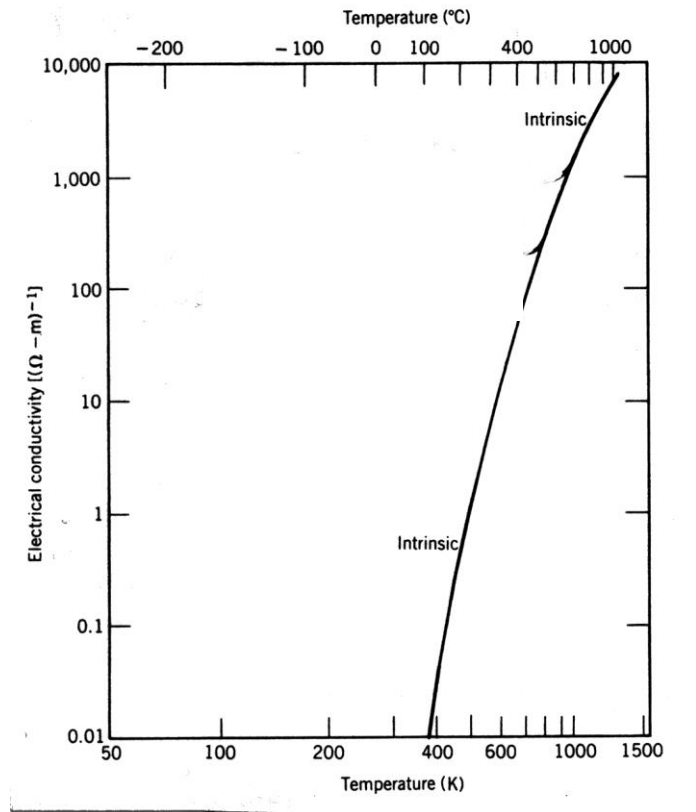
INTRINSIC SEMICONDUCTORS

Q4. Justify the different effects that an increase on the exposure temperature have on a metal and on an intrinsic semiconductor

metal



intrinsic semiconductor



INTRINSIC SEMICONDUCTORS

Intrinsic semiconductors: elemental materials

Pure elements of group **IVA** in the periodic table: **Si**, **Ge** $\Rightarrow$ “covalent bonds”

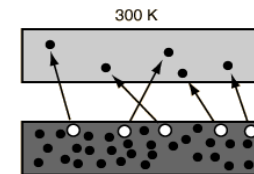
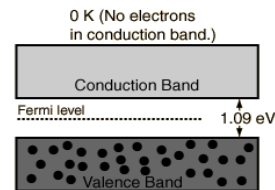
As the atom size increases



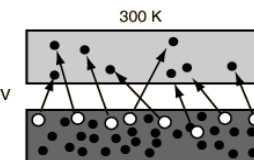
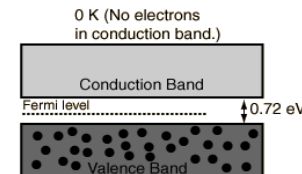
E_g decreases



σ increases for intrinsic semiconductors



SILICON



GERMANIUM

	Silicon	Germanium
Energy gap, eV	1.1	0.67
Electron mobility μ_n , $\text{m}^2/(\text{V} \cdot \text{s})$	0.135	0.39
Hole mobility μ_p , $\text{m}^2/(\text{V} \cdot \text{s})$	0.048	0.19
Intrinsic carrier density n_i , carriers/ m^3	1.5×10^{16}	2.4×10^{19}
Intrinsic resistivity ρ , $\Omega \cdot \text{m}$	2300	0.46
Density, g/m^3	2.33×10^6	5.32×10^6

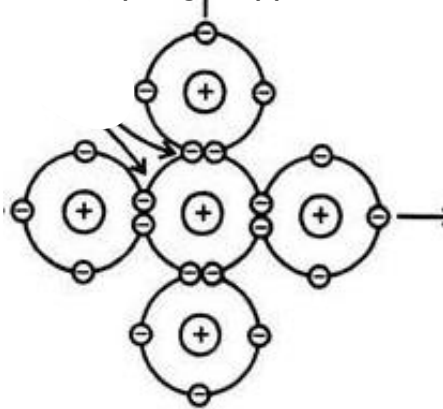
INTRINSIC SEMICONDUCTORS

Intrinsic semiconductors: Intermetallic materials

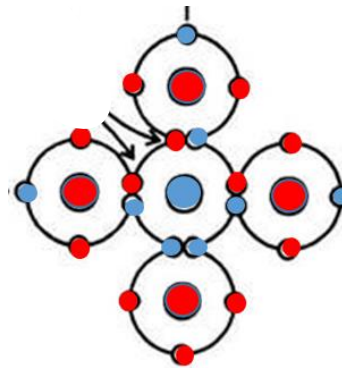
Compounds between elements of groups IIIA y VA: **GaAs, InSb**

Compounds between elements of groups IIB y VIA: **CdS, ZnTe**

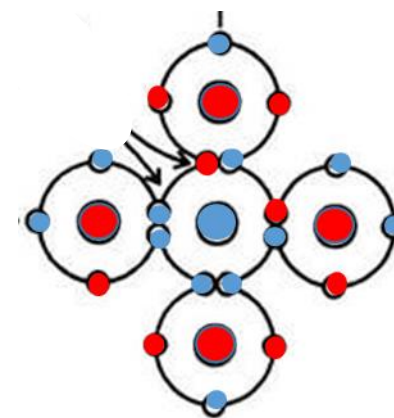
Elemental semiconductor
(14 group)



Compound semiconductor
(13-15 groups)



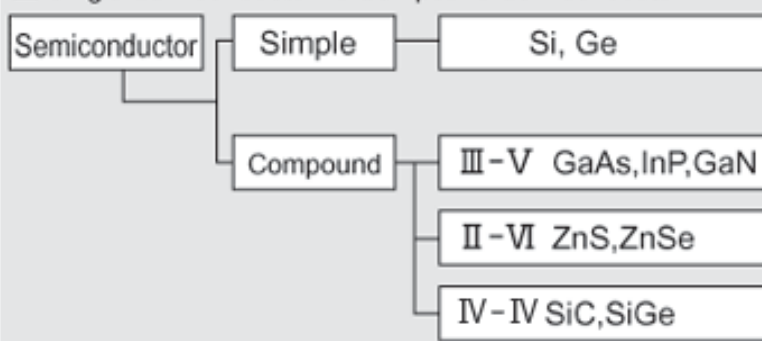
Compound semiconductor
(12-16 groups)



+covalent

+ionic

■ Single Semiconductor vs Compound Semiconductor



■ Combinations on the Periodic Table of the Elements

II	III	IV	V	VI
	B	C	N	
	Al	Si	P	
Zn	Ga	Ge	As	Se
Cd	In	Sn	Sb	Te

As covalent character decreases

E_g increases

σ decreases
for **intrinsic** semiconductors

INTRINSIC SEMICONDUCTORS

For intrinsic semiconductors, the number of charge carriers is reduced

The high chemical and structural purity favors high mobilities



The conductivity is determined by n

<i>Material</i>	<i>Band Gap (eV)</i>	<i>Electrical Conductivity [($\Omega\cdot m$)⁻¹]</i>	<i>Electron Mobility (m²/V-s)</i>	<i>Hole Mobility (m²/V-s)</i>
Elemental				
Si	1.11	4×10^{-4}	0.14	0.05
Ge	0.67	2.2	0.38	0.18
III-V Compounds				
GaP	2.25	—	0.03	0.015
GaAs	1.42	10^{-6}	0.85	0.04
InSb	0.17	2×10^4	7.7	0.07
II-VI Compounds				
CdS	2.40	—	0.03	—
ZnTe	2.26	—	0.03	0.01

In **intrinsic semiconductors**: $\mu_e > \mu_h$

Q10. Order the following semiconductors according to their conductivity (assume that the materials are always at the same temperature). Explain the criterion you have followed

- **AlP, GaAs, and InSb**

<i>Material</i>	<i>Band Gap (eV)</i>	<i>Electrical Conductivity [($\Omega\cdot m$)⁻¹]</i>	<i>Electron Mobility (m²/V-s)</i>	<i>Hole Mobility (m²/V-s)</i>
Elemental				
Si	1.11	4×10^{-4}	0.14	0.05
Ge	0.67	2.2	0.38	0.18
III-V Compounds				
GaP	2.25	—	0.03	0.015
GaAs	1.42	10^{-6}	0.85	0.04
InSb	0.17	2×10^4	7.7	0.07
II-VI Compounds				
CdS	2.40	—	0.03	—
ZnTe	2.26	—	0.03	0.01

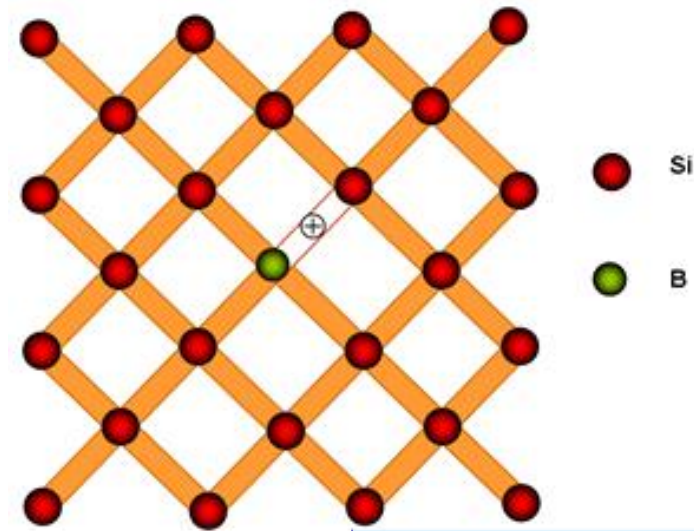
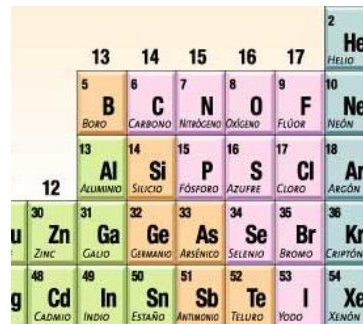
■ Combinations on the Periodic Table of the Elements

II	III	IV	V	VI
	B	C	N	
	Al	Si	P	
Zn	Ga	Ge	As	Se
Cd	In	Sn	Sb	Te

2 types depending on the doping atoms

Type - p

Less valence e^-
than the doped material



EXTRINSIC SEMICONDUCTORS - TYPE n

Impurity atoms have more valence electrons than the atoms in the network → behave as **electron donors** (Group VA) P, As, Sb

Impurities with higher valence

DONOR impurities

Excess of e^-

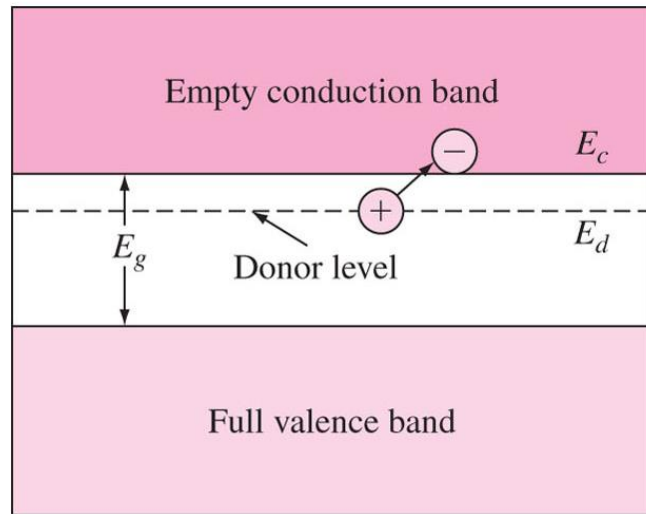
e^- in the conduction band are the majority charge carriers

$$\sigma \approx n_e \cdot |q_e| \cdot \mu_e \quad (n_e \gg n_H)$$

Electrons at the donor energy level E_d jump into the conduction band ($\Delta E = E_c - E_d$) → small amount of energy

Distance from the donor level to the conduction band → determines n_e

Copyright © The McGraw-Hill Companies, Inc. All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or by any information storage and retrieval system, without prior written permission from The McGraw-Hill Companies, Inc.



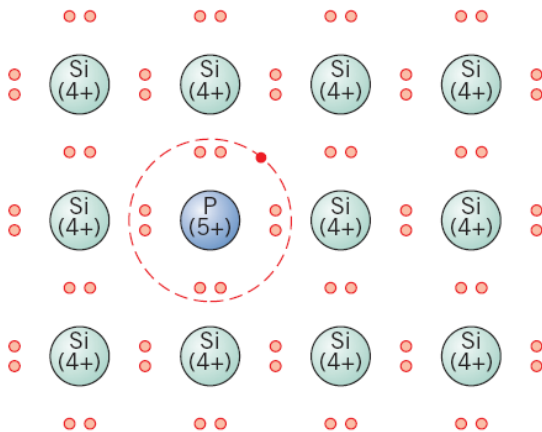
$$(\Delta E = E_c - E_d)$$

Groups 4A and 5A
of periodic table

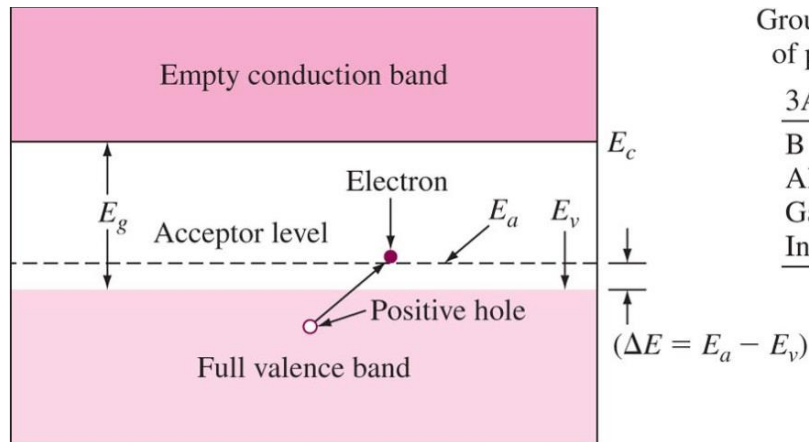
4A	5A
C	N
Si	P
Ge	As
Sn	Sb

$Si \rightarrow 4e^-$ valence

$P \rightarrow 5e^-$ valence



EXTRINSIC SEMICONDUCTORS - TYPE p



Groups 3A and 4A
of periodic table

3A	4A
B	C
Al	Si
Ga	Ge
In	Sn

Impurity atoms have less valence electrons than the atoms in the network → behave as acceptors

B, Al, Ga (group VA)

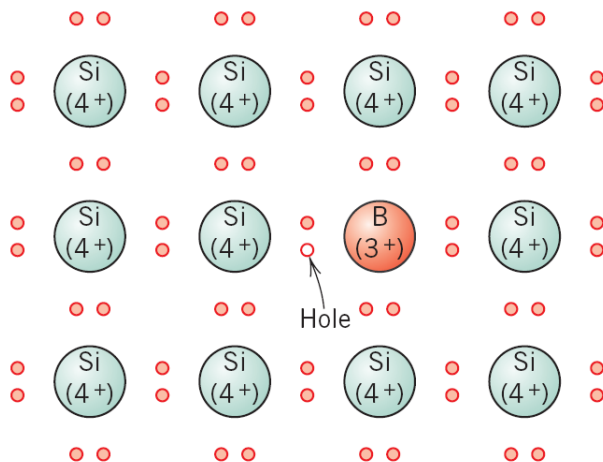
↓
Impurities with lower valence

↓
ACCEPTORS impurities

↓
Excess of holes

$Si \rightarrow 4e^-$ valence

$B \rightarrow 3e^-$ valence

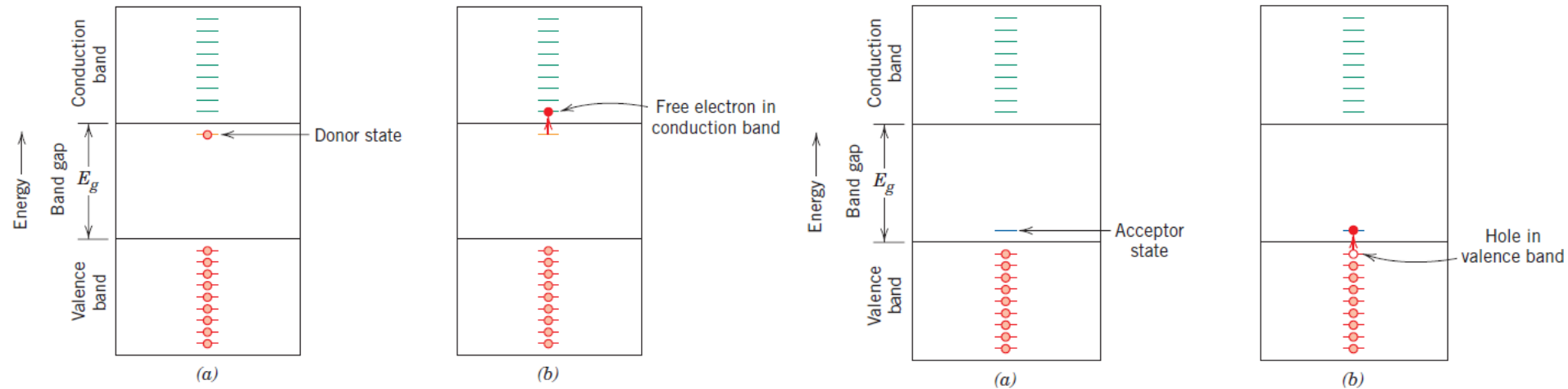


Holes moving through the **valance band** are the majority charge carrier

$$p \gg n \Rightarrow \sigma \cong p \cdot |q_e| \mu_h$$

Acceptor energy level E_a located within the band gap and just above the top of the valence band → small amount of energy ($\Delta E = E_a - E_v$) for excitation of an electron into the acceptor level, leaving behind a hole in the valence band.

EXTRINSIC SEMICONDUCTORS



Excitation from a donor state in which a free electron is generated in the conduction band.

Excitation of an electron into the acceptor level, leaving behind a hole in the valence band.

Semiconductor	Majority-carrier concentrations	Minority-carrier concentrations
n-type	n_n (concentration of electrons in n-type material)	p_n (concentration of holes in n-type material)
p-type	p_p (concentration of holes in p-type material)	n_p (concentration of electrons in p-type material)

EXTRINSIC SEMICONDUCTORS

Q5. Justify whether the following extrinsic semiconductors are p-type or n-type and whether the introduced impurities work as donors or as acceptors:

- a- **Silicon doped with antimony** in substitutional solid solution
- b- **Germanium doped with arsenic** in substitutional solid solution
- c- **Silicon doped with gallium** in substitutional solid solution

The periodic table shows elements color-coded by groups: Group 1 (yellow), Group 2 (orange), Groups 3-10 (green), Groups 11-12 (light blue), Groups 13-16 (pink), Group 17 (red), Group 18 (light blue), and Lanthanides/Actinides (purple). Arrows indicate the placement of impurities: a blue arrow from Group 15 (Antimony) points to Silicon (Group 14), a red arrow from Group 15 (Arsenic) points to Germanium (Group 14), and a green arrow from Group 13 (Gallium) points to Silicon (Group 14).

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H																	He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	Lr	Rf	Db	Sg	Bh	Hs	Mt									
6			La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb		
7			Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No		

EXTRINSIC SEMICONDUCTORS

	Dopant	Silicon		Germanium	
		E_d	E_a	E_d	E_a
Ionic radius ↓	P	0.045	Increasing σ	0.0120	
	As	0.049		0.0127	
	Sb	0.039		0.0096	
Ionic radius ↓	B		0.045		0.0104
	Al		0.057	Increasing σ	0.0102
	Ga		0.065		0.0108
	In		0.160		0.0112

The donor and acceptor energy gaps (in electron volts) when silicon and germanium semiconductors are doped.

Element	Dopant	Periodic table group of dopant	Maximum solid solubility of dopant (atoms/m ³)
Si	B	IIIA	600×10^{24}
	Al	IIIA	20×10^{24}
	Ga	IIIA	40×10^{24}
	P	VA	1000×10^{24}
	As	VA	2000×10^{24}
	Sb	VA	70×10^{24}
Ge	Al	IIIA	400×10^{24}
	Ga	IIIA	500×10^{24}
	In	IIIA	4×10^{24}
	As	VA	80×10^{24}
	Sb	VA	10×10^{24}

Some extrinsic, elemental semiconductors

Q10. Order the following semiconductors according to their conductivity (assume that the materials are always at the same temperature). Explain the criterion you have followed

- Germanium with 5 ppm of As Germanium with 5 ppm of Sb.
- Silicon with 10 ppm of B Silicon with 10 ppm of In.
- Germanium with 5ppm of As Germanium with 5 ppm of Sb.

■ Combinations on the Periodic Table of the Elements

II	III	IV	V	VI
	B	C	N	
	Al	Si	P	
Zn	Ga	Ge	As	Se
Cd	In	Sn	Sb	Te

	Dopant	Silicon		Germanium	
		E_d	E_a	E_d	E_a
Ionic radius ↓	P	0.045	Increasing σ	0.0120	Increasing σ
	As	0.049		0.0127	
	Sb	0.039		0.0096	
Ionic radius ↓	B		0.045		0.0104
	Al		0.057		0.0102
	Ga		0.065		0.0108
	In		0.160		0.0112

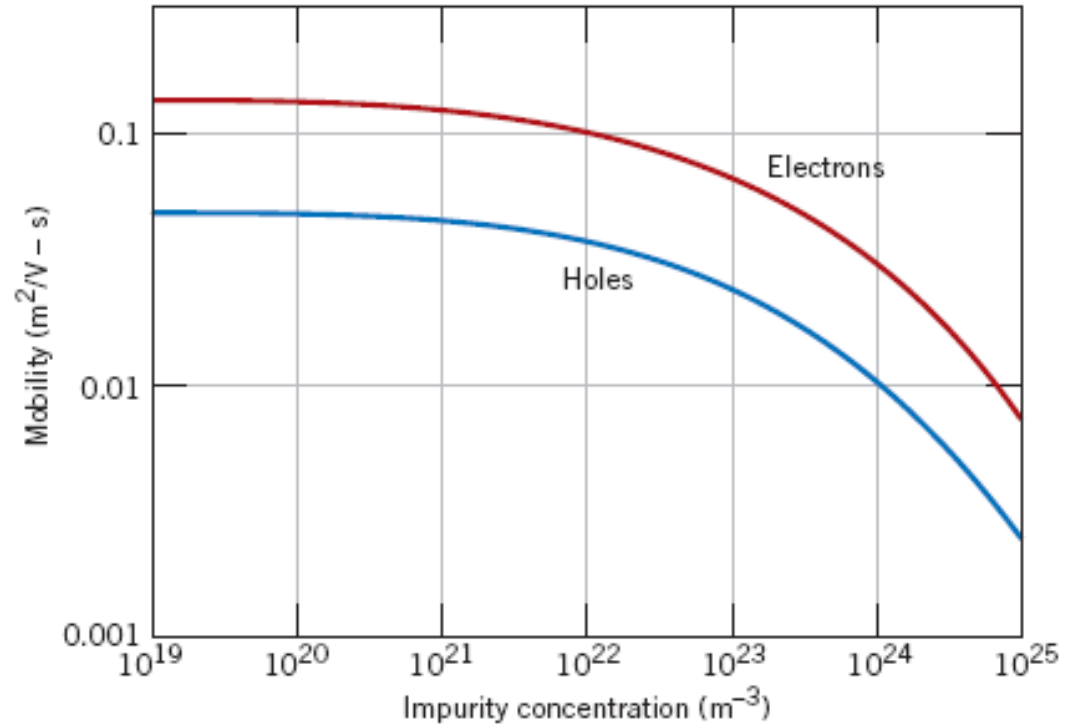
The donor and acceptor energy gaps (in electron volts) when silicon and germanium semiconductors are doped.

EXTRINSIC SEMICONDUCTORS

MOBILITY OF CHARGE CARRIERS

In extrinsic semiconductors: $\mu_e > \mu_h$

For extrinsic semiconductors the distortion of the crystallographic structure caused by doping is negligible, so σ is controlled by the number of charge carriers



Dependence of room temperature electron and hole mobilities on dopant concentration for Si

Semiconductor	μ_e / μ_h
Si	2,81
Ge	2,05

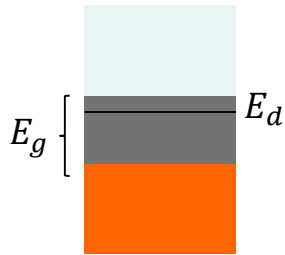
EXTRINSIC SEMICONDUCTORS

EFFECT OF TEMPERATURE IN EXTRINSIC SEMICONDUCTORS

At low T: number of ionized impurities / volume, determines σ

When T increases $\rightarrow$ no. Ionized atoms increases $\rightarrow \sigma$ increases $\rightarrow$ semiconductor extrinsic range (slope of the line depends on the type of semiconductor)

n type

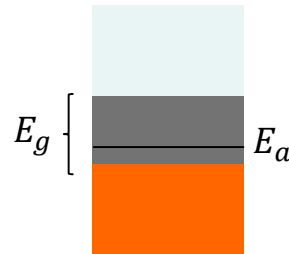


$$\sigma = \sigma_0 \exp\left(-\frac{E_c - E_d}{2kT}\right)$$

$$\ln(\sigma) = \ln(\sigma_0) - \frac{E_c - E_d}{2kT}$$

Slope n type: $-\frac{E_c - E_d}{2kT}$

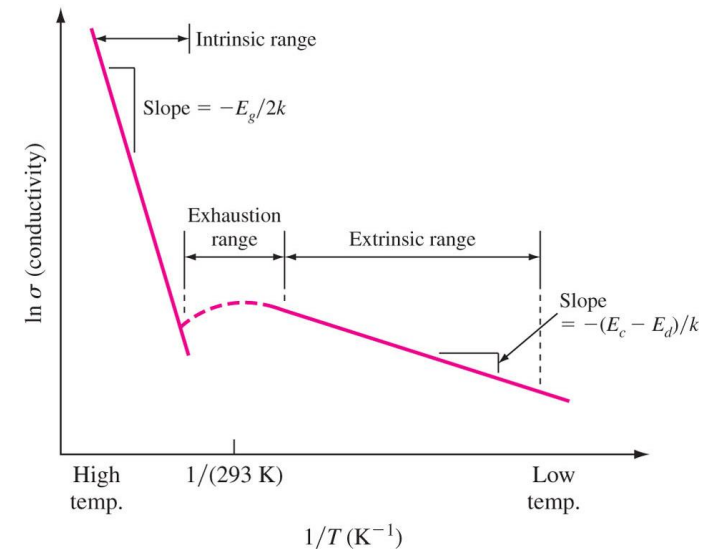
p type



$$\sigma = \sigma_0 \exp\left(-\frac{E_a - E_c}{2kT}\right)$$

$$\ln(\sigma) = \ln(\sigma_0) - \frac{E_a - E_c}{2kT}$$

Slope p type: $-\frac{E_a - E_c}{2kT}$



At intermediate T: all the impurities have been ionized: σ is constant with T

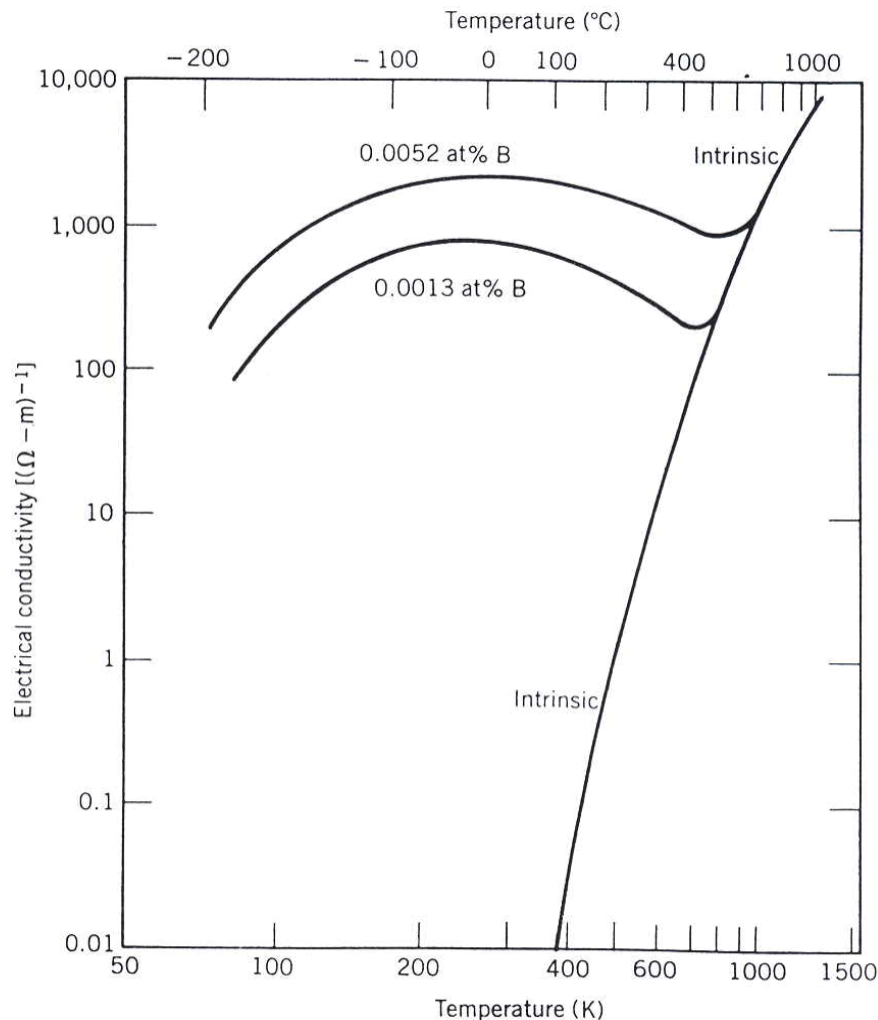
- **Type n:** all the electrons are lost $\rightarrow$ exhaustion range
- **Type p:** all atoms ionized $\rightarrow$ saturation range

At high T: intrinsic region

EXTRINSIC SEMICONDUCTORS

EFFECT OF TEMPERATURE IN EXTRINSIC SEMICONDUCTORS

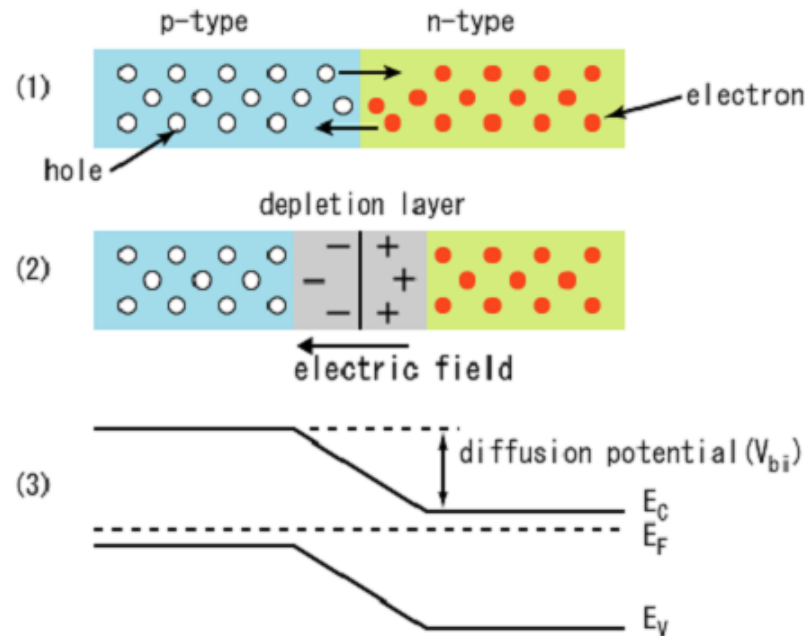
For **extrinsic** semiconductors:



- **Low and Moderate T:** The material behaves as an extrinsic conductor and the conductivity is governed by the number of ionized atoms.
- **High T:** The conductivity slightly decreases because the number of charge carriers remains constant and the disordering effect of temperature becomes evident.
- **Very high T:** The material behaves as an intrinsic conductor

APPLICATIONS OF SEMICONDUCTORS

p–n junction is a boundary or interface between two types of semiconductor material, p-type and n-type,...



<https://www.youtube.com/watch?app=desktop&v=btOIDQeMrMg>

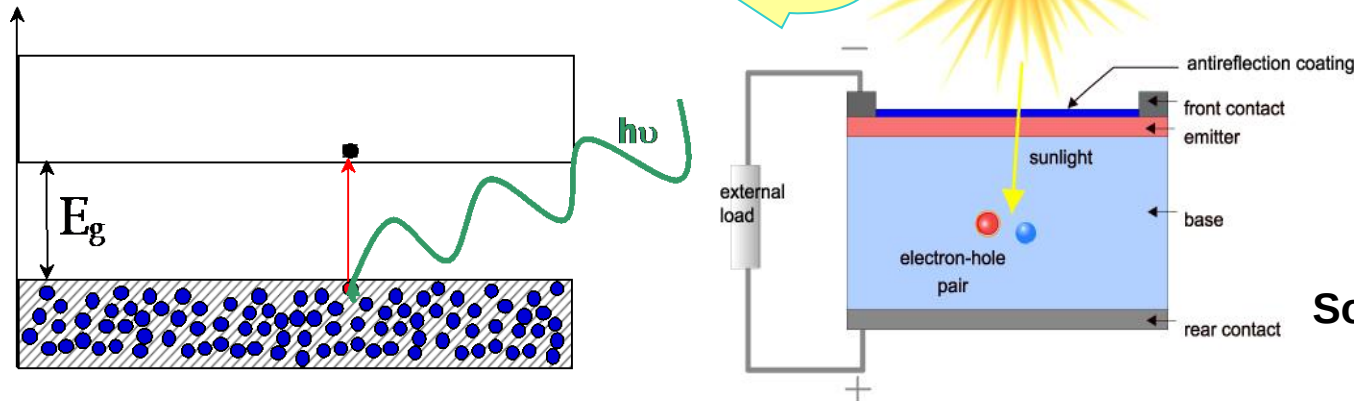
https://www.youtube.com/watch?v=ar7xDMR4P_U

APPLICATIONS OF SEMICONDUCTORS

e- jump from valence to conduction band:

- Due to thermal excitation
- Due to intense electrical fields
- Due to exposure to radiation with a given λ

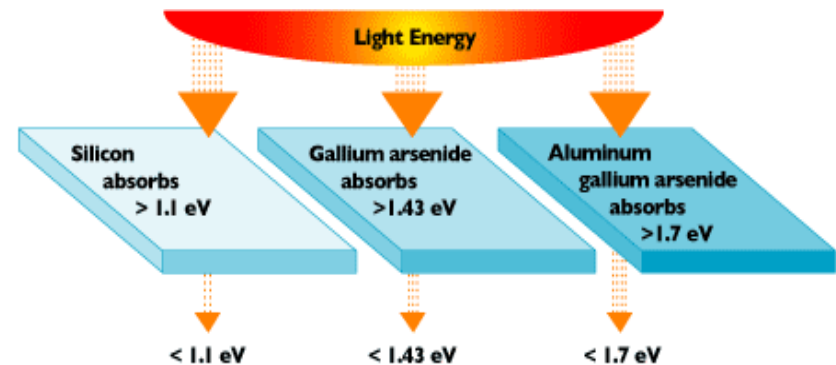
Photoconductivity



Solar light $E_g \sim 1.5 \text{ eV}$

$\sigma \rightarrow$ Photovoltaic cells

Photoconductivity is a phenomenon in which a material becomes more conductive due to the absorption of electro-magnetic radiation such as visible light,



APPLICATIONS OF SEMICONDUCTORS

LED (Light Emitting Diode): Semiconductor doped to create a p-n bond that emits monochromatic light when it is directly polarized.

Colour depends on the distance between valence and conduction band

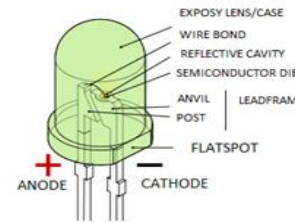
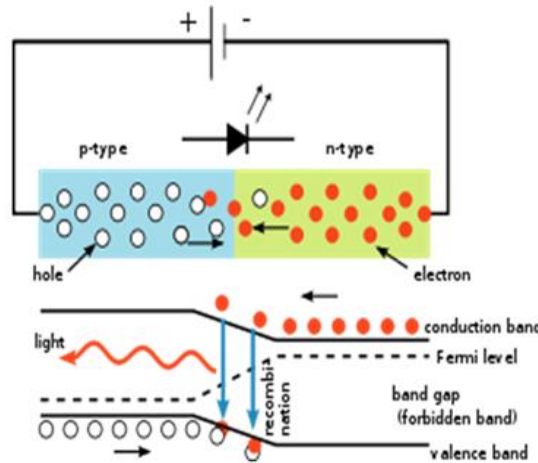


Diagram of LED

Material	Radiation
AlGaAs	red and infra-red
GaN	green
GaP	green
ZnSe	blue
InGaN	blue
SiC	blue
Diamond	ultra-violet

E
g

λ



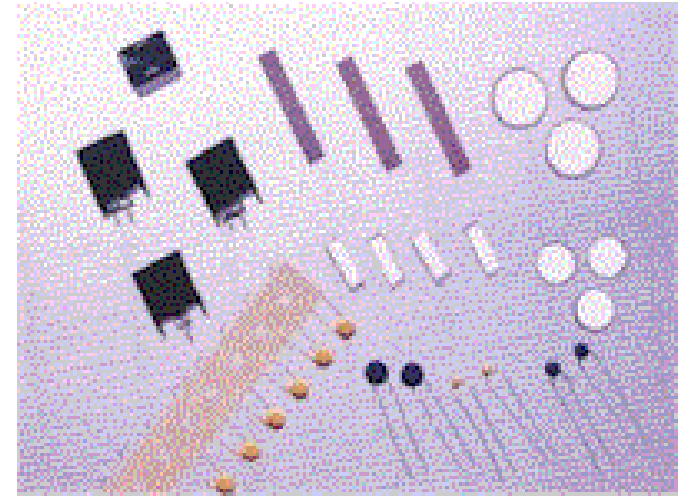
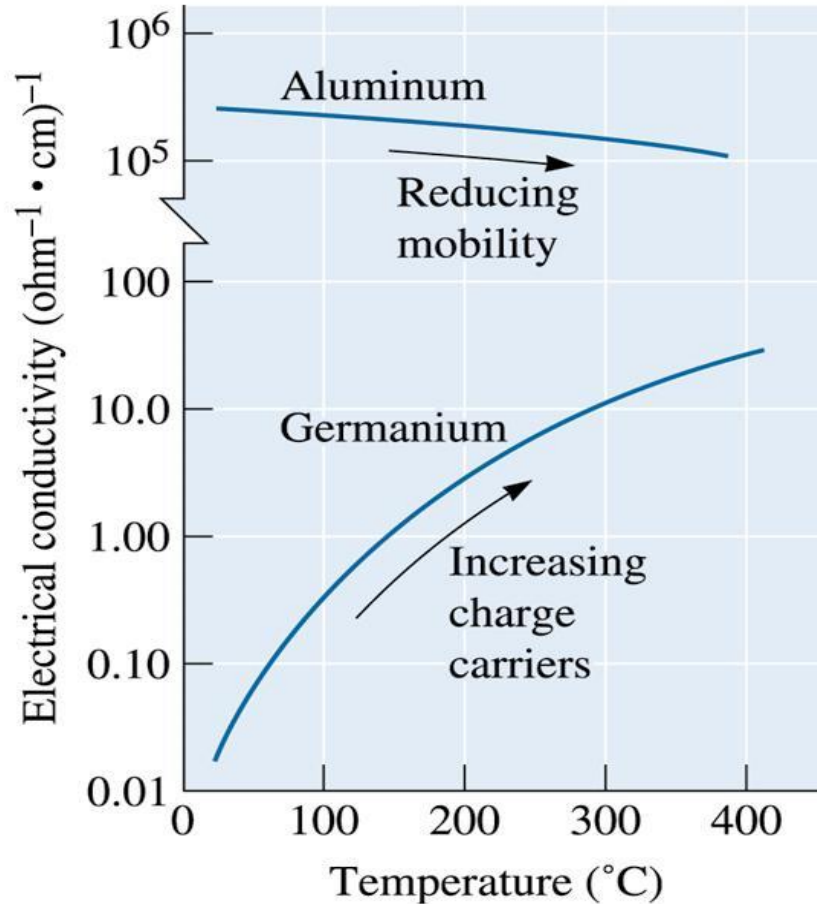
Wikipedia

APPLICATIONS OF SEMICONDUCTORS

THERMISTORS

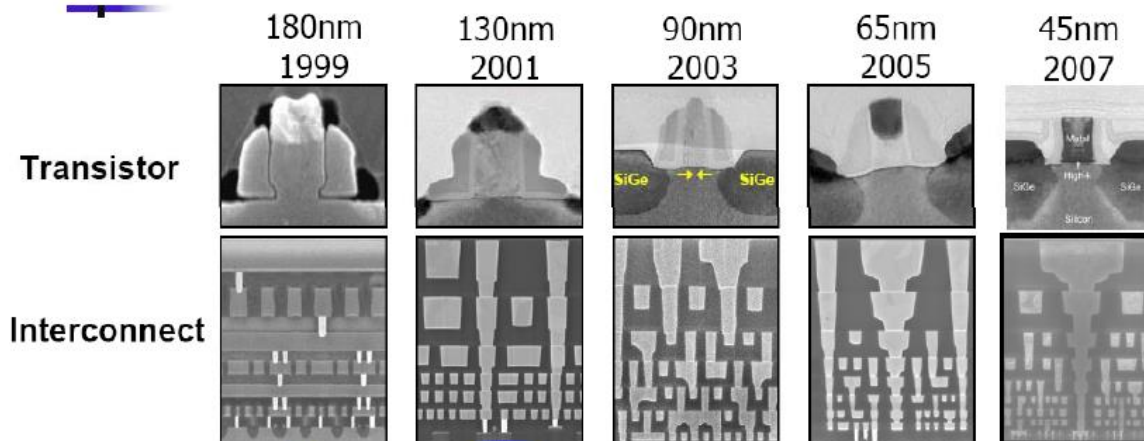
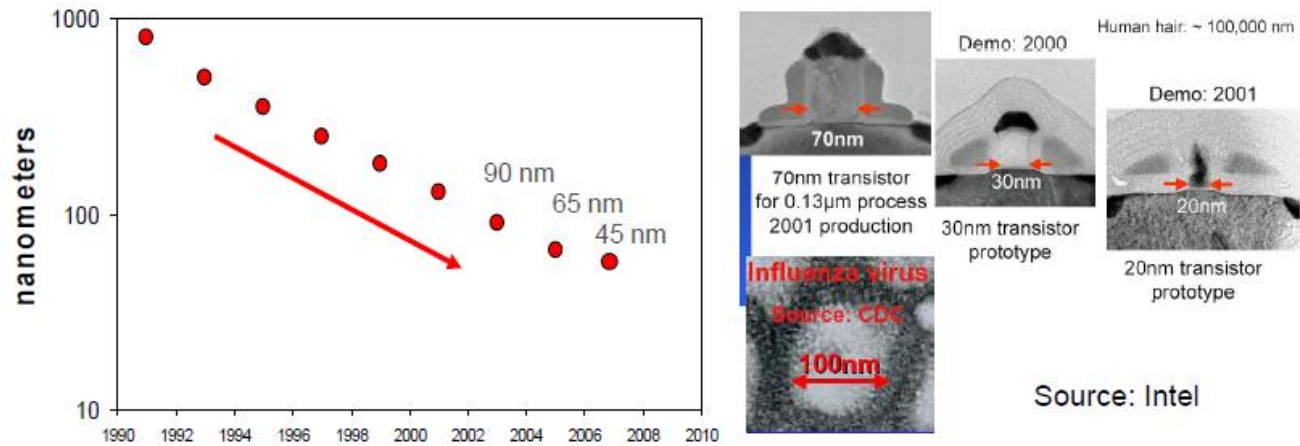
Devices based on the change of electrical conductivity of semiconductors with temperature.

©2003 Brooks/Cole, a division of Thomson Learning, Inc. Thomson Learning is a trademark used herein under license.



APPLICATIONS OF SEMICONDUCTORS

Transistor - A semiconductor device that can be used to amplify electrical signals.



TOPIC 6. ELECTRICAL PROPERTIES OF MATERIALS

Band structure and electrical properties

Conducting materials

Metallic conductors

Non-metallic conductors

Semiconducting materials

Intrinsic

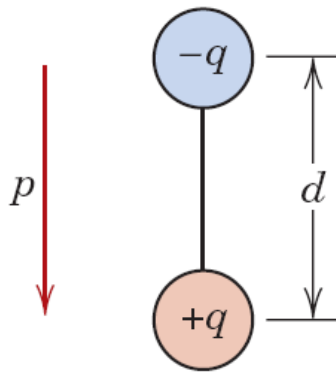
Extrinsic (n and p)

Insulating and Dielectric materials

INSULATING AND DIELECTRIC MATERIALS

Electrically insulating material (**high E_g**) that exhibits –or could be made to exhibit- a **dipolar structure**

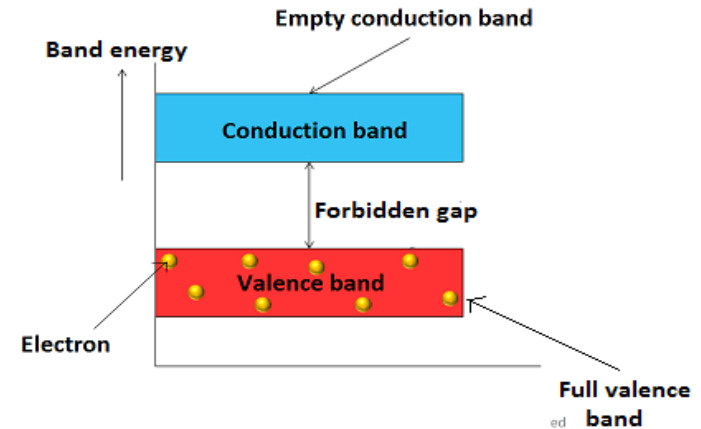
Dipole moment



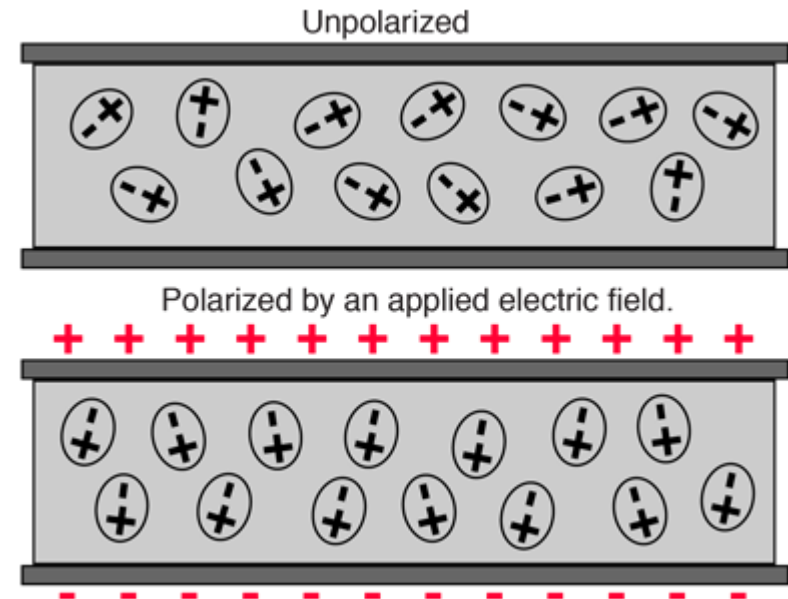
$$\text{Dipole moment} = p = q \cdot d$$

q = dipole charge

d = distance of separation between the positive and negative ends of dipole



Polarization

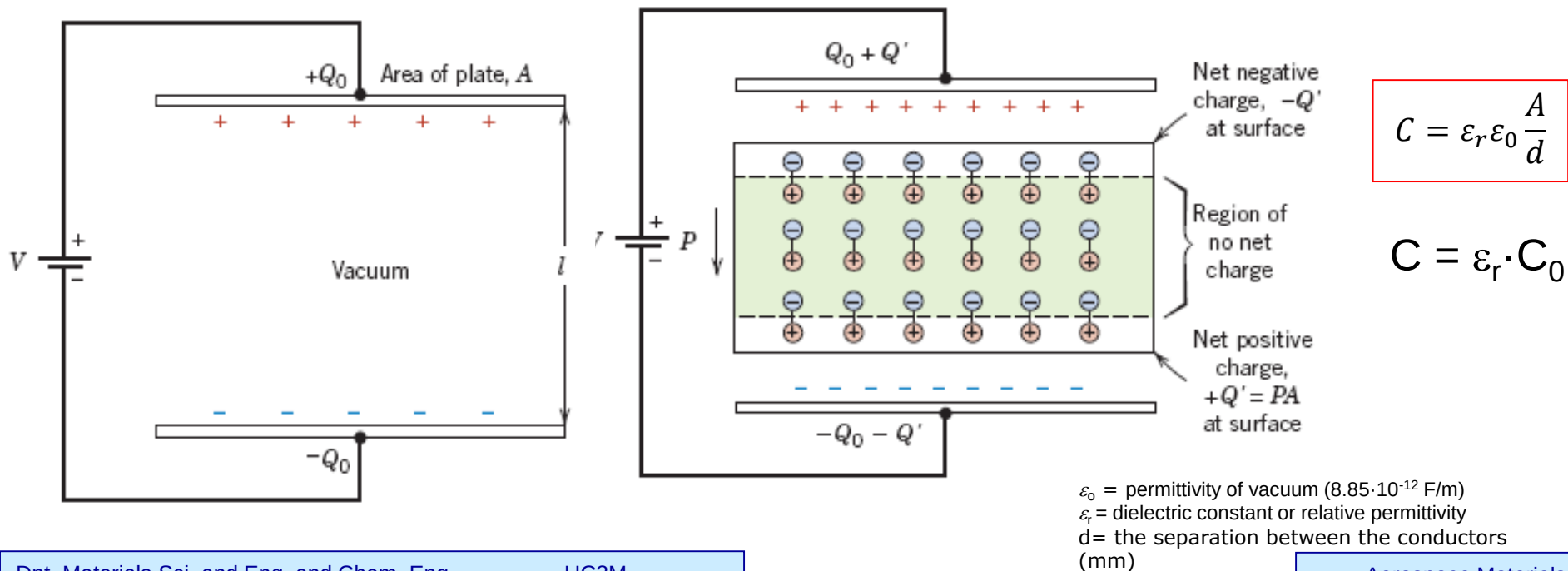


INSULATING AND DIELECTRIC MATERIALS

Application of highly-resistive materials

Depending on their response to the electric field:

- **Insulators** (Insulate conductors from each other and from a metal mass)
→ High E_g
- **Dielectric materials** (Between the plates of a capacitor, increases its capacitance)
→ Intense dipoles that easily line up to the electric field



INSULATING AND DIELECTRIC MATERIALS

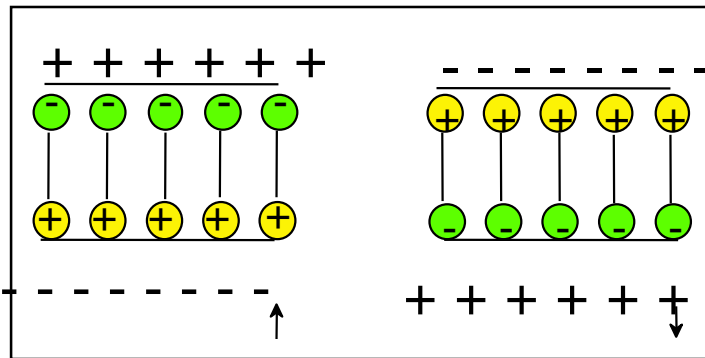
Loss factor ($\tan \delta$)

Capacitors are normally used in alternating current circuits



Dipoles must realign fast when the electric field changes

$\tan \delta$ = Amount of electrical energy **lost**, such as **heat**, by a capacitor in an alternating current circuit, due to friction among dipoles when they tried to reorient



Increase with: **-Moisture**
-Temperature
- Frequency of the current

↓ resistivity

Dielectric Strength

Maximum electric field that the dielectric can maintain without electrical breakdown

$$E_{break} = \frac{U_{break}}{d}$$

d= the separation between the conductors (mm)

INSULATOR AND DIELECTRIC MATERIALS

Insulator materials should have:

High Resistivity (ρ). To avoid current leaks

High Dielectric Strength E_{break} To avoid insulate breakdowns

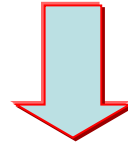
Loss Factor ($\tan \delta$) Small

Low Permittivity (ϵ) To avoid polarization

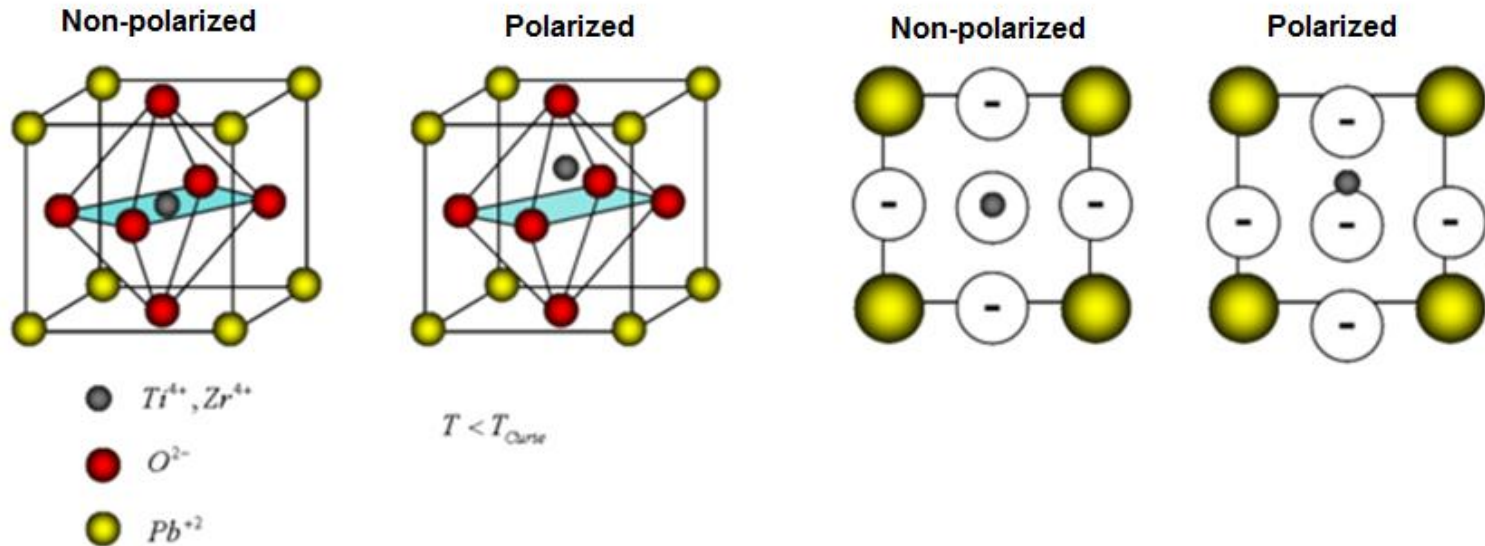
Material	Volume Resistivity ($\Omega \cdot m$)	Dielectric Strength		Dielectric Constant K		Loss Factor	
		V/mil	kV/mm	60 Hz	10^6 Hz	60 Hz	10^6 Hz
Electrical porcelain insulators	10^{11} - 10^{13}	55-300	2-12	6	...	0.06	...
Steatite insulators	$>10^{12}$	145-280	6-11	6	6	0.008-0.090	0.007-0.025
Fosterite insulators	$>10^{12}$	250	9.8	...	6	...	0.001-0.002
Alumina insulators	$>10^{12}$	250	9.8	...	9	...	0.0008-0.009
Soda-lime glass	...	...	...	...	7.2	...	0.009
Fused silica	...	8	...	...	3.8	...	0.00004

FERROELECTRIC MATERIALS

Ferroelectric materials have permanent dipoles that can remain aligned in the absence of an electric field



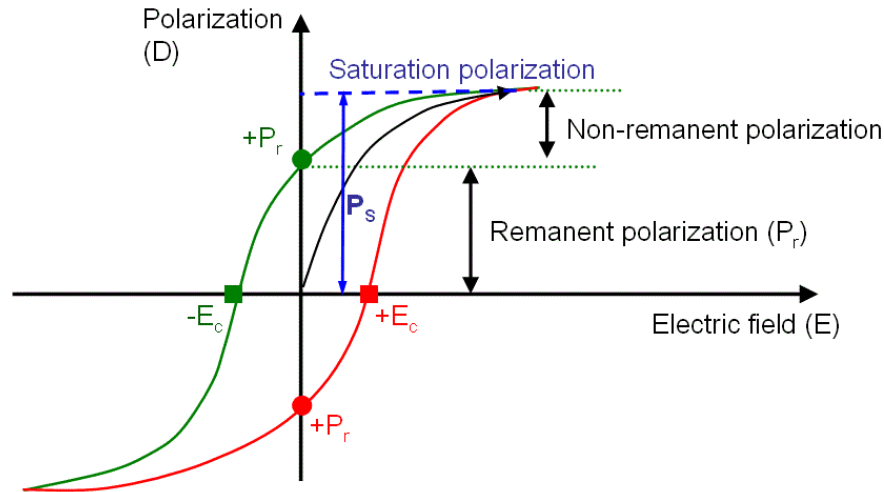
spontaneous polarization in the absence of external electric fields



If a ferroelectric is slowly cooled to temperatures lower than the critical temperature, in the presence of a strong electric field, the **anions** can be displaced to the **field positive side**, and the **cations** to the **field negative side**. It produces a permanent electric dipole.

FERROELECTRIC MATERIALS

Electric field hysteresis cycles are similar to the ones in ferromagnetic materials with magnetic field



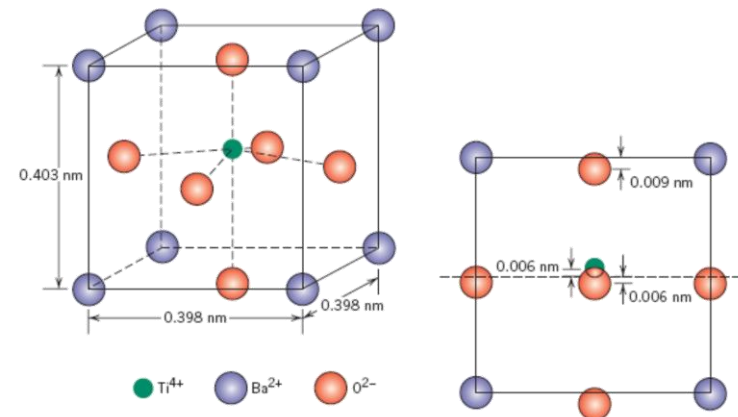
The area under the hysteresis curve is related to the energy required to change the orientation of the dipoles in the material

Curie Temperature: Temperature above which the material loses its ferroelectric properties and becomes dielectric

$$-200\text{ }^{\circ}\text{C} < T_c < 640\text{ }^{\circ}\text{C} \text{ (typically)}$$

Example: BaTiO_3 has $T_c = 120^{\circ}\text{C}$

- Under $120\text{ }^{\circ}\text{C}$ the structure is tetragonal
- Above 120°C the crystalline structure of BaTiO_3 is cubic and the permanent dipoles disappear



FERROELECTRIC MATERIALS

Material		ϵ_r
Ferroelectric	BaTiO ₃	4000
	KNbO ₃	700
	Rochelle salt	170
Dielectric	TiO ₂	100-50
	Al ₂ O ₃	10
	Porcelain	6.0
	PET	3.2
	PVC	3,5
	Polyethylene	2.3
	Air	1.0

$\epsilon_r(\text{ferroelectric}) \gg \epsilon_r(\text{common dielectric}) \Rightarrow$ Small and very efficient capacitors (for example, for computer memories)

PIEZOELECTRIC

Piezoelectric Materials = **ferroelectric materials** that can be polarized when external mechanical stress is applied.

*It occurs in **ceramics** with a crystal structure with a **low degree of symmetry***

Direct piezoelectric effect

External stress



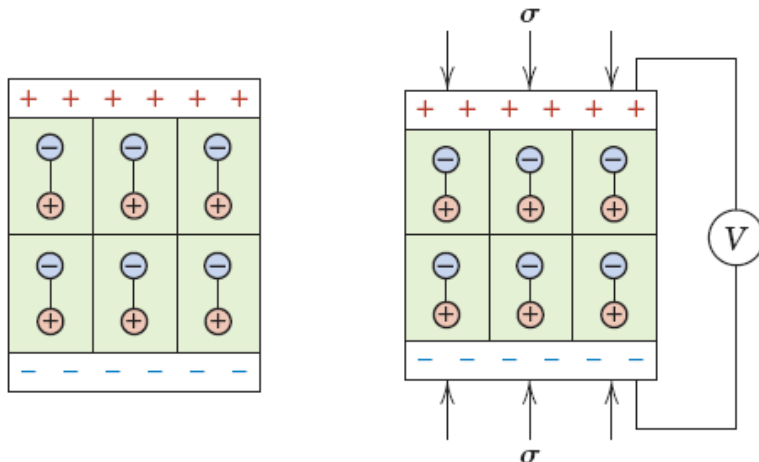
Crystalline structure deformation



Dipole alteration



Electric field generation



Reverse piezoelectric effect (electrostriction)

External electric field



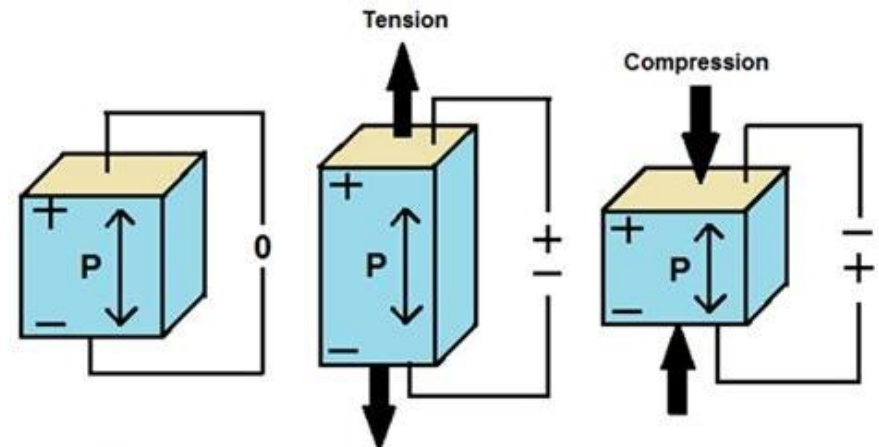
Dipole alteration

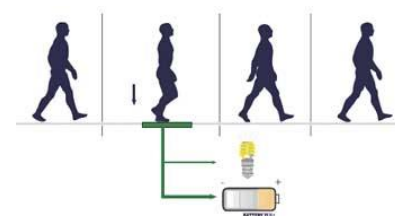


Crystalline structure deformation



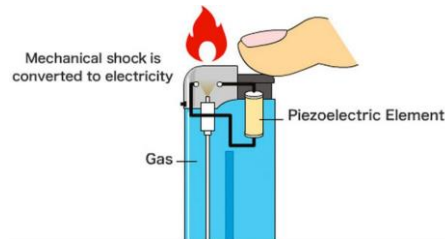
Dimensional alteration in the material



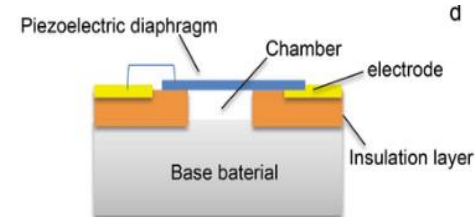
[illegible]

PIEZOELECTRICS

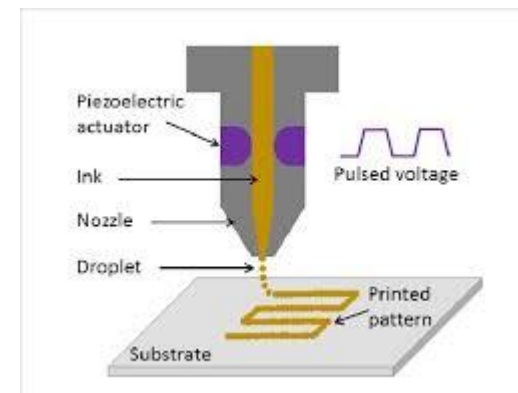
Lighters



Transducers in speakers



Inkjet printers



Microphones, Hydrophones

